CHRONO-Fair: Anytime-Valid Counterfactual Flip Monitoring for Clinical Machine Learning

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Clinical prediction models are deployed under regulatory regimes that require fairness monitoring after release. The FDA 2024 Predetermined Change Control Plan guidance and EU AI Act Articles 10 and 17 both expect continuous post-market bias surveillance. The auditing tools in current use compute a group fairness metric on a fixed snapshot of predictions. They then report a point estimate against a threshold, such as the 0.8 rule. A point estimate, however, hides whether the verdict it produces is stable. On the real Texas-100X discharge dataset (925 128 records), the worst-group true-positive-rate verdict for race reads "fair" at a point value of 0.824. The same verdict flips to "unfair" in 4 of 20 hospital-network resamples, a Verdict Flip Rate of 0.20. Existing clinical fairness-audit tools generally do not report when such a verdict flips as patients arrive, nor whether the flip is driven by biased data or by insufficient data.

The reason this matters is that the three tools available to a governance team each address only part of the problem, and in isolation. A bootstrap confidence interval reveals that a verdict is uncertain, but it is computed once and is silent during deployment. The Verdict Flip Rate quantifies how often the verdict flips across resamples, but it is a scalar with no time axis and no statistical false-alarm control. Drift detectors such as ADWIN operate sequentially, but they monitor a single metric and provide no anytime-valid guarantee under repeated inspection. None of the three attributes a detected flip to a cause that a clinical governance team can act on.

This paper presents CHRONO-Fair, a deployment-oriented statistical monitoring framework. It extends static verdict-flip auditing into time-resolved monitoring of counterfactual verdict flips, and combines two evidence layers on a single dataset. The first layer is a real Texas-100X batch verdict-stability analysis. The second is a set of Texas-100X-calibrated controlled streaming experiments for sequential monitoring behaviour. The framework has four coordinated estimators. A Flip Hazard survival object recasts the scalar flip rate as a group-conditional Kaplan-Meier curve with log-rank tests. A per-cell e-process gives anytime-valid Type-I control under repeated inspection through Ville's inequality. A step-wise Benjamini-Hochberg adjustment is then applied across the intersectional cells at each inspection time. This adjustment is a cross-sectional correction, not a full time-uniform online false-discovery guarantee. An ensemble entropy decomposition splits a detected flip into an aleatoric (data) cause and an epistemic (sample-size) cause. A Wasserstein-1 Regression Counterfactual Allocation Parity statistic extends the analysis to length-of-stay regression. On the real Texas-100X verdict data, the confidence interval and the Verdict Flip Rate agree on all 12 audited metric-attribute combinations, and both isolate the single unstable verdict that the point estimate misses. On Texas-100X-calibrated controlled streams, the e-process detects an injected drift in 50 of 50 runs at a median delay of 828 patients, against 21 of 50 for ADWIN, while holding the empirical false-alarm rate at or below the nominal level. All streaming findings are controlled replay or simulation evidence, not prospective clinical validation, which remains future work.

Additional Key Words and Phrases: algorithmic fairness, clinical machine learning, anytime-valid inference, fairness drift, post-deployment monitoring, length-of-stay prediction

1 INTRODUCTION

Healthcare prediction models can pass a one-off fairness audit at the time of deployment and develop subgroup gaps later. Obermeyer et al. [83] documented racial bias in a widely used care-management algorithm. Seyyed-Kalantari et al. [99] reported underdiagnosis of female and minority patients by chest radiograph classifiers. Habib et al. [44] and Wong et al. [115] showed that the Epic Sepsis Model underperformed in external validation. Davis et al. [29] analysed eleven years of VA cohort data and observed that fairness gaps drift after deployment, frequently in the direction of widening.

These post-deployment failures have shifted regulatory attention from one-off model validation toward lifecycle monitoring. The U.S. FDA codified this shift in its Predetermined Change Control Plan final guidance of December 2024, which asks manufacturers to specify how a model will be monitored for bias after release [104]. The EU AI Act goes further: Article 17, binding from August 2027 for high-risk medical AI, requires continuous post-market bias monitoring [37]. The reporting standards moved in parallel, with DECIDE-AI [106], TRIPOD+AI [26], and CONSORT-AI [69] all now expecting lifecycle rather than snapshot evaluation. Taken together with the argument of McCradden et al. [76] and Eaneff et al. [36] for "algorithmic stewardship" as a regulatory category, the direction of travel is clear: fairness is expected to be an observed property of a deployed system, not a property checked once.

The methodological toolchain available to satisfy that expectation, however, remains snapshot-based. The open-source audit packages in routine use, Aequitas, Fairlearn, AIF360, and the Microsoft Responsible AI Dashboard, all operate in batch mode on a fixed set of predictions. The online fair-learning line of work [50, 51, 82] does monitor a stream, but it has been demonstrated on COMPAS and credit data rather than on a clinical cohort. Anytime-valid inference [43, 48, 93, 114] supplies the sequential statistical machinery, yet it has been applied to A/B-test monitoring and calibration drift, not to fairness drift at intersectional granularity. The predictive-multiplicity literature [13, 27, 74] characterises pipeline instability, but it frames that instability as a reliability concern and stops short of a fairness verdict. No existing line of work, in short, monitors the thresholded fairness verdict itself as patients arrive.

The closest precedent is the Verdict Flip Rate (VFR), introduced in prior work as a scalar that quantifies how often a fairness verdict flips between resamples of a hospital-network bootstrap [7]. VFR captures what the canonical group metrics miss, namely the instability of a fairness verdict under stochastic pipeline choices, and it does so on the object hospitals act on, the pass-or-fail verdict. It is, however, a static and batch quantity: a single scalar at a single threshold that gives no indication of when, where, or why a gap arises during deployment. The framework presented here, CHRONO-Fair, extends VFR along exactly that missing axis. Where VFR-Audit measures static batch verdict reliability, CHRONO-Fair monitors counterfactual verdict flips as the patient stream unfolds, and Figure 1 contrasts the two.

Contributions. This paper makes four contributions, each presented as a simulation-based prototype.

(1) A Flip Hazard estimator that recasts the resampling-based VFR as a group-conditional survival object on the patient arrival index. Kaplan-Meier survival, Nelson-Aalen cumulative hazard, two-sample log-rank statistics, and a Restricted Mean Flip Time (RMFT) are reported per protected group.
(2) A per-cell anytime-valid e-process monitor for the flip indicator, with a shrunk GROW betting rule and a warm-up of 100 patients. A step-wise Benjamini-Hochberg adjustment is applied across intersectional cells at each inspection time. The empirical Type-I rate stays at or below the nominal α on the calibrated streams.
(3) An ensemble entropy decomposition that splits the per-cell flip mass into aleatoric and epistemic components. The split admits a clear governance interpretation: an

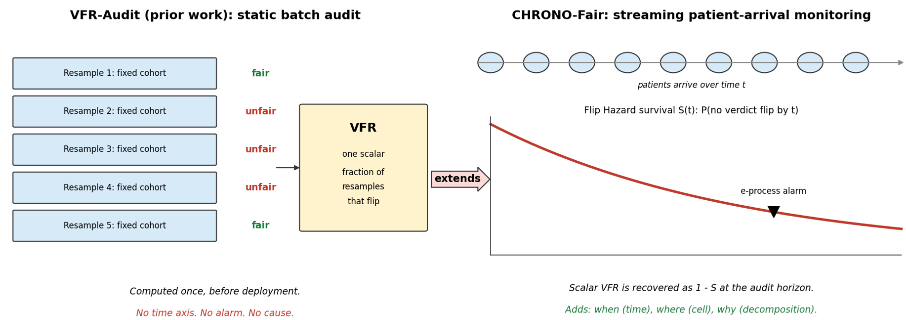

Fig. 1. VFR-Audit is a static batch scalar. CHRONO-Fair is the streaming extension. The scalar Verdict Flip Rate is recovered as $1 - S$ at the audit horizon, and the framework adds the time axis, the alarm, and the cause.

aleatoric-dominated cell signals label or feature-pipeline bias, an epistemic-dominated cell signals sample-size scarcity.

(4) A Regression Counterfactual Allocation Parity (RCAP) metric for length-of-stay regression. RCAP is the Wasserstein-1 distance between counterfactual rank distributions and is sensitive to allocation shifts that the group-MAE comparator does not register.

To our knowledge, no single published system combines these four estimators with intersectional anytime-valid control and an automated inspection report. The framework, the synthesiser, and the dashboard prototype are released as open-source code with the supplementary material.

2 RELATED WORK

2.1 Group, individual, and counterfactual fairness

Dwork et al. [35] defined individual fairness as a Lipschitz constraint. Two individuals within distance d on a task metric must receive outcome distributions within distance D. Hardt et al. [45] defined equalised odds as equal true-positive and false-positive rates across groups. Equal opportunity relaxes this to equal true-positive rate only. Chouldechova [24] proved that, when group base rates differ, a classifier cannot hold predictive parity, equal false-positive rate, and equal false-negative rate at once. Kleinberg et al. [63] proved a parallel result: calibration and class-balanced error are jointly attainable only under equal base rates or perfect prediction. Pleiss et al. [88] showed the conflict persists for any non-trivial calibration constraint and quantified the achievable relaxation. Kusner et al. [65] defined counterfactual fairness as invariance of the prediction distribution under a do-intervention on the protected attribute in a structural causal model. Chiappa [23] restricted this to unfair causal paths only, leaving resolving paths intact. Nabi and Shpitser [79] constrained path-specific effects during estimation by a constrained likelihood. Kilbertus et al. [61] separated proxy discrimination from explanatory effects in a causal graph. Makhlouf et al. [71] mapped 19 causal fairness notions to the data assumptions each one needs. The surveys of Verma and Rubin [107], Mehrabi et al. [77], Pessach and Shmueli [86], and Caton and Haas [19] catalogue 20 or more metrics. They report that the choice of metric, not the model, drives most reported disparity.

On mitigation, Agarwal et al. [2] reduced fair classification to a sequence of cost-weighted subproblems solved by a saddle-point game. Donini et al. [32] added a fairness term to empirical risk minimisation with a generalisation bound. Calmon et al. [17] and Kamiran and Calders [57] reweighted or transformed training data before fitting. Madras et al. [70] and Lahoti et al. [66] learned representations that an adversary cannot use to recover the protected attribute. Wang et al. [112] bounded fairness violation when group labels are noisy. Kallus et al. [56] estimated disparity by combining a main dataset with an auxiliary dataset that carries the protected attribute.

2.2 Subgroup, multicalibration, and statistical-audit perspectives

Kearns et al. [59] showed that parity on coarse groups can hide large violations on structured subgroups, an effect they named fairness gerrymandering. Hébert-Johnson et al. [47] defined multicalibration as calibration that holds simultaneously on every subgroup in a computationally identifiable class. Kim et al. [62] produced multiaccuracy by post-processing, auditing for subgroups where the residual correlates with subgroup membership. Hashimoto et al. [46], Sagawa et al. [95], and Duchi and Namkoong [33] minimised worst-group loss through distributionally robust optimisation over an uncertainty set of group mixtures. Martínez et al. [73] characterised the accuracy-fairness frontier as a Pareto set and computed it through a multi-objective descent. Cherian and Candès [22] produced finite-sample confidence intervals for a fairness metric using bootstrap and rank-based statistics. Besse et al. [11] reported that the sampling variance of statistical parity can exceed the disparity itself when subgroup size is below a few hundred. Coston et al. [28] evaluated counterfactual risk under selective labels through inverse-propensity weighting. These works audit a fixed snapshot. None updates a fairness verdict sequentially as patients arrive.

2.3 Predictive multiplicity, model multiplicity, and stability

Breiman [16] observed that distinct models can attain near-identical loss while disagreeing on individuals. Marx et al. [74] formalised this as predictive multiplicity. They defined ambiguity as the fraction of inputs that receive a conflicting prediction from some model within an ϵ-loss Rashomon set. Watson-Daniels et al. [113] extended the measure to probabilistic outputs through the range of predicted scores. Semenova et al. [98] linked a large Rashomon set to the existence of a simpler accurate model. Dong and Rudin [31] computed variable-importance ranges across the Rashomon set rather than for one model. Black et al. [13] argued that model multiplicity gives a designer latitude to select a fairer model at no accuracy cost. Cooper et al. [27] showed that a measured group gap can be a variance artefact: averaging predictions across the Rashomon set shrank reported disparity. Miller and Blume [78] defined the empirical Decision Flip Rate as the fraction of patients whose binary decision changes between bootstrap re-trainings. Black et al. [14] detected individual discrimination through an optimal-transport map between group-conditional input distributions. The Verdict Flip Rate of [7] measured how often the fairness verdict, not the prediction, flips between hospital-network resamples.

2.4 Uncertainty quantification and its decomposition

Depeweg et al. [30] decomposed predictive uncertainty into two terms. Aleatoric uncertainty is the mean of per-model entropies. Epistemic uncertainty is the entropy of the mean prediction minus that mean, equal to the mutual information between label and parameters. Hüllermeier and Waegeman [49] formalised the distinction and noted that only the epistemic term shrinks with more data. Lakshminarayanan et al. [67] estimated epistemic uncertainty from the disagreement of M independently trained networks. Gal and Ghahramani [40] approximated the same quantity through Monte Carlo dropout. Kendall and Gal [60] separated learned input-dependent variance from model uncertainty in a single loss. Ovadia et al. [84] reported that deep ensembles kept calibration error lowest among uncertainty methods under distribution shift. Alghamdi et al. [4] applied the aleatoric-epistemic split to discrimination, attributing the irreducible part to the data and the reducible part to the model class. Romano et al. [94] and Gibbs and Candès [42] produced prediction sets with a finite-sample coverage guarantee, the latter adapting the set width online under shift. Barber et al. [8] extended conformal coverage to non-exchangeable data with a bound that degrades in the total-variation distance from exchangeability. Angelopoulos and Bates [6] surveyed the coverage guarantees that the conformal family provides. Dutta et al. [34] and Calmon et al. [18] decomposed discrimination through partial information, separating the exempt component from the unfair component.

2.5 Anytime-valid inference, drift detection, online learning

Ville [108] proved that a non-negative supermartingale started at one crosses $1/\alpha$ with probability at most α. This single inequality underwrites every e-process. Wald [111] built the sequential probability ratio test, which stops as soon as a likelihood ratio leaves a fixed interval. Howard et al. [48] produced confidence sequences, intervals that cover the parameter simultaneously at every sample size with probability $1-\alpha$. Waudby-Smith and Ramdas [114] estimated a bounded mean by betting. The wealth process $\prod_t(1+\lambda_t(X_t-m))$ is a martingale under the null mean m. The GROW rule picks λ_t to maximise expected log-growth. Ramdas et al. [93] and Shafer [100] unified these results as game-theoretic statistics, where evidence equals the capital won against the null. Grünwald et al. [43] characterised the e-variable that is growth-optimal against a composite alternative. Johari et al. [54] applied always-valid p-values to continuous A/B-test monitoring, removing the penalty for repeated peeking. Podkopaev and Ramdas [89] tracked a deployed model's risk through a betting martingale on the loss sequence. Foster and Stine [39] introduced alpha-investing, which spends a false-discovery budget across an ordered hypothesis stream. Javanmard and Montanari [52] produced the LORD rules that control online false-discovery rate with a decaying memory. Bifet and Gavalda [12] introduced ADWIN, which keeps a variable-length window and drops the older half when two sub-window means differ by a Hoeffding bound. Gama et al. [41] taxonomised concept drift into sudden, gradual, and recurring types. Rabanser et al. [91] reported that dimensionality reduction followed by a two-sample test detected dataset shift most reliably. Iosifidis and Ntoutsi [50, 51] maintained a fairness constraint online by reweighting the loss as a stream evolves. None of these monitors a counterfactual flip indicator under intersectional multiple-testing control.

2.6 Clinical machine learning fairness, prediction, and deployment

Obermeyer et al. [83] traced the bias of a care-management algorithm to its label. The model predicted healthcare cost as a proxy for healthcare need. At a fixed risk score, Black patients carried more chronic conditions than White patients. Correcting the label raised the share of Black patients flagged for extra care from 17.7% to 46.5%. Vyas et al. [109] catalogued race-adjusted clinical formulas, including an eGFR correction that lowered estimated kidney disease severity for Black patients. Seyyed-Kalantari et al. [99] reported that chest radiograph classifiers had higher false-negative rates for female, Black, and publicly insured patients, with the largest underdiagnosis on intersectional subgroups. Wong et al. [115] externally validated the Epic Sepsis Model. They found an area under the receiver operating characteristic curve of 0.63, against the vendor-reported 0.76 to 0.83. Habib et al. [44] confirmed the external-validation gap and the high alert burden. Noseworthy et al. [81] tested an electrocardiogram deep model and found stable performance across race once age and sex were matched. Larrazabal et al. [68] showed that a sex-imbalanced training set lowered diagnostic accuracy for the under-represented sex. Coley et al. [25] reported that a suicide-risk model had lower sensitivity for Black and American Indian patients. Adam et al. [1] found that race-correlated language in clinical notes shifted downstream model recommendations. Pierson et al. [87] trained a pain model on patient-reported scores rather than radiologist labels and explained a larger share of the racial pain gap. Davis et al. [29] tracked an in-use model across eleven years of VA data and observed subgroup performance drift after deployment. Rajkomar et al. [92] and Chen et al. [20] set out fairness desiderata for clinical machine learning. Singh et al. [101] argued that a deployed clinical model needs continual rather than one-off evaluation. Finlayson et al. [38] catalogued the dataset-shift modes that degrade clinical models, including case-mix and practice-pattern change. Vasey et al. [106] and Collins et al. [26] issued the DECIDE-AI and TRIPOD+AI reporting standards, both of which now require subgroup performance reporting. Sculley et al. [97] and Breck et al. [15] documented the monitoring debt that accrues when a model ships without a test harness. The MIMIC-IV [55] and eICU [90] databases supply the public critical-care streams used for external validation. Texas-100X [53] provides 925 128 hospital discharge records with the demographic mix this paper's synthesiser reproduces.

2.7 Counterfactual auditing of deployed classifiers

Maughan and Near [75] proposed prediction sensitivity, a per-batch scalar. It estimates how often a counterfactual swap of the protected attribute changes the verdict, without the attribute at inference time. Wachter et al. [110] defined a counterfactual explanation as the smallest input change that flips the decision. Ustun et al. [105] certified whether such a change is attainable through actionable features only. Chen et al. [21] bounded a disparity estimate when the protected attribute is unobserved at audit time. These methods return a scalar or a per-instance explanation at a fixed time. They provide no anytime-valid alarm and no decomposition of the flip into a data cause and a model cause. CHRONO-Fair targets that gap.

2.8 Positioning against prior work

Table 1 places CHRONO-Fair against eight families of prior work on nine capability axes. The axes are the ones a post-deployment clinical monitor needs. Static or streaming records the operating mode. Verdict reliability records whether the method reports the stability of a pass or fail verdict. Anytime-valid per-cell monitoring records a false-alarm rate bounded under inspection at every step. Cross-cell adjustment records a multiple-testing correction across monitored cells. Intersectional records cells defined by two or more protected attributes. Counterfactual-flip monitoring records tracking of verdict flips under a protected-attribute counterfactual. Root-cause output records attribution of a detected gap to a data or a model cause. Regression support records handling of a continuous, rank-allocation output. Real clinical evidence records the empirical clinical data used.

Read row by row, the table shows that the gap is structural rather than incidental, since no prior method family holds more than two of the nine capabilities at once. The nearest entries each fall short on a different axis. Anytime-valid risk tracking [89] is streaming and anytime-valid, but it monitors aggregate loss rather than a group-conditional fairness verdict and offers no diagnostic attribution. Fairness-drift monitoring [29] is the closest

Table 1. Capability comparison of fairness-audit method families. "Y" denotes the capability is present as a designed feature, "–" denotes absent or not reported, and "P" denotes partial support. The final row is the contribution of this paper; the footnote states what each "P" on that row means.

Method family (representative work)	Static/ streaming	Verdict reliability	Anytime-valid per-cell	Cross-cell adjustment	Inter- sectional	CF-flip monitoring	Root- cause	Regression/ rank-alloc.	Real clinical evidence
Group metrics, batch toolkits [45]	static	–	–	–	P	–	–	–	various
Bootstrap-CI fairness audit [22]	static	Y	–	–	–	–	–	–	–
Predictive multiplicity, eDFR [74, 78]	static	Y	–	–	–	–	–	–	some
Verdict Flip Rate [7]	static	Y	–	–	P	Y	–	–	Texas-100X batch
Prediction sensitivity [75]	per batch	–	–	–	–	Y	–	–	–
Online fair learning [50]	streaming	–	–	–	–	–	–	–	–
Fairness-drift monitoring [29]	quarterly	P	–	–	P	–	–	–	VA cohort
Anytime-valid risk tracking [89]	streaming	–	Y	–	–	–	–	–	–
CHRONO-Fair (this paper)	**streaming**	**Y**	**Y**	**P**	**Y**	**Y**	**P**	**Y**	**P**

Footnote. For the CHRONO-Fair row, "P" on cross-cell adjustment means an inspection-time step-wise Benjamini-Hochberg adjustment rather than a full time-uniform online false-discovery procedure. "P" on root-cause means a diagnostic triage signal that separates aleatoric from epistemic flip mass, not a confirmed causal root cause. "P" on real clinical evidence means a real Texas-100X batch verdict-stability analysis combined with Texas-100X-calibrated controlled streaming and replay experiments, not prospective deployment evidence.

clinical work, yet it operates on a quarterly cadence, monitors a single metric, and gives no statistical guarantee under repeated inspection. The Verdict Flip Rate [7] measures verdict instability directly but collapses it to a scalar with no time axis. CHRONO-Fair occupies the remaining position by combining the streaming, anytime-valid, intersectional, and diagnostic capabilities, while the three partial marks in its row record exactly where its support is qualified rather than complete.

2.9 Open gaps addressed by this paper

Four gaps follow from Table 1. *Gap 1, the time axis.* A batch verdict cannot say when a fairness gap opened during deployment, and a clinical response needs that timestamp. *Gap 2, the peeking penalty.* A governance team inspects a monitor often. A fixed-sample test inflates its false-alarm rate under repeated looks. A continuous monitor therefore needs an anytime-valid guarantee. *Gap 3, the cause.* A detected gap with no attributed cause does not tell a team whether to fix the data pipeline or to collect more samples. *Gap 4, regression outputs.* Length-of-stay and cost models produce continuous outputs. The flip-rate and verdict literature is built for binary classification. CHRONO-Fair addresses the four gaps with the Flip Hazard estimator, the e-process, the aleatoric-epistemic decomposition, and RCAP, in that order.

3 PROBLEM FORMULATION

3.1 Clinical fairness, the fairness verdict, and the flip indicator

This paper uses three terms with fixed meaning throughout. *Clinical fairness* denotes the property that the thresholded decisions of a deployed clinical machine learning model do not depend on a patient's legally or ethically protected attributes beyond what is clinically justified, evaluated by an agreed group-fairness metric and threshold (for example disparate impact above 0.8). The *fairness verdict* is the binary pass-or-fail outcome of that evaluation at a given threshold. The *counterfactual verdict flip*, or *verdict flip* for short, is the event in which the verdict changes between the model's factual decision on a patient and its counterfactual decision under a swap of the protected attribute. The framework treats the verdict flip as the observable governance object and monitors its rate as patients arrive.

3.2 Notation and assumptions

Let $\{(X_i, A_i, Y_i)\}_{i\geq 1}$ be a stream of patient records, where $X_i \in \mathbb{R}^d$ are non-sensitive features, A_i is a protected attribute taking values in a finite set $\mathcal{A}$, and $Y_i \in \{0,1\}$ or $Y_i \in \mathbb{R}_{\geq 0}$ is the label. The arrival index t orders patients by admission time. The deployed model at index t is $f_t\colon (X, A) \to \hat{Y}$. The model may change between two indices through periodic retraining, threshold recalibration, or hot-fixes. Re-training events are exposed to CHRONO-Fair as version stamps. Between two retrains, the model is held fixed and the framework monitors the decision stream rather than the model parameters.

Let A_i^{cf} denote a counterfactual sensitive-attribute value drawn from $\mathcal{A}$. In this paper, A_i^{cf} is fixed to a reference level (typically the majority group). The counterfactual prediction $\hat{Y}_i^{\mathrm{cf}} = f_t(X_i, A_i^{\mathrm{cf}})$ is obtained by setting the protected attribute to its counterfactual value and removing the marginal effect on observable features through a feature-shift adjustment. This corresponds to the deployment-time counterfactual of Maughan and Near [75]. It is weaker than the SCM-based counterfactual of Kusner et al. [65] but it is computable at inference time without an audited causal graph.

3.3 Flip indicator

For binary classification with threshold τ, the flip indicator is

$$Z_i = \mathbf{1}\Big\{\mathbf{1}\Big\{\hat{Y}_i \geq \tau\Big\} \neq \mathbf{1}\Big\{\hat{Y}_i^{\mathrm{cf}} \geq \tau\Big\}\Big\}. \tag{1}$$

For regression, a threshold-free analogue is the counterfactual rank shift formalised in Section 4.4. The flip indicator depends on f_t at the patient's arrival index, not on f_t at the reporting time, so retraining events do not retroactively invalidate prior Z_i.

3.4 Baseline calibration

Each cell c defined by a value of the protected attribute has a baseline flip rate $\rho_0^{(c)}$. The baseline is calibrated on the pre-deployment validation set used to lock the model for release. The validation set should match the deployment population and contain at least 500 patients per cell. If the per-cell calibration size falls below 500, $\rho_0^{(c)}$ is set to the marginal flip rate on the same validation set and a wider warm-up window is used. Calibration is repeated each time the model is replaced.

4 METHOD

CHRONO-Fair is one framework with four coordinated estimators that share a single counterfactual flip indicator. Figure 2 shows the data flow. A patient stream feeds the deployed model and its counterfactual, the flip indicator is computed, and the four estimators consume it. The Flip Hazard estimator and the e-process are sequential. The decomposition and RCAP run on demand. The Inspector report aggregates the four outputs for governance review.

4.1 Flip Hazard estimator

Definition 4.1 (Flip Hazard). For protected group a, the Flip Hazard function is

$$\lambda_a(t) = \lim_{\Delta\to 0} \frac{1}{\Delta} \Pr(Z_t = 1, T \geq t \mid A = a) \big/ \Pr(T \geq t \mid A = a), \tag{2}$$

where $T = \inf\{t : Z_t = 1\}$ is the first flip time and the limit is taken on the patient-arrival index.

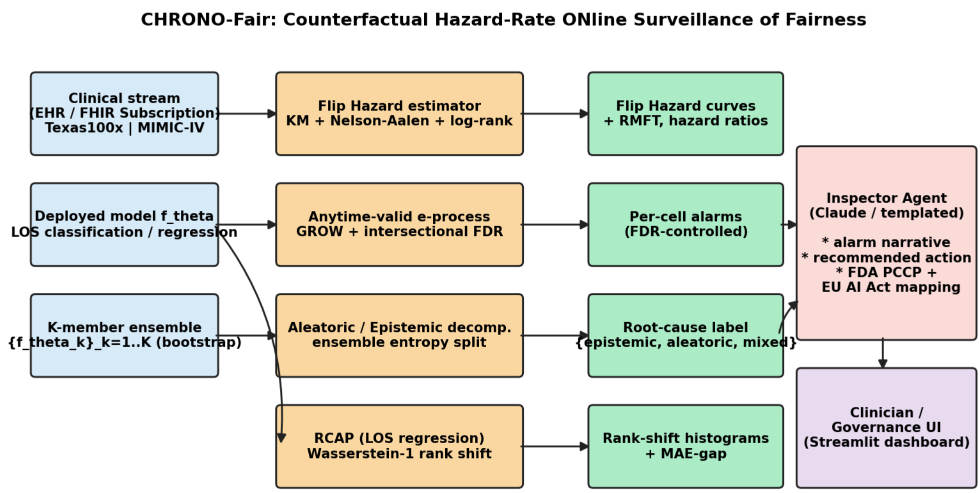

Fig. 2. CHRONO-Fair architecture. A clinical stream feeds a deployed model and a bootstrap ensemble. Four estimators run concurrently: Flip Hazard survival, anytime-valid e-process, aleatoric/epistemic decomposition, and RCAP. The Inspector Agent translates alarms into a structured governance report.

The no-flip survival is $S_a(t) = \exp(-\int_0^t \lambda_a(s)\,ds)$. The scalar VFR of [7] corresponds to the aggregate $1 - S_a(\tau^\star)$ at the full audit horizon $\tau^\star$. The framework reports survival as a curve, not a scalar.

The estimator is non-parametric, so it makes no distributional assumption on the flip-time. Kaplan-Meier survival [58] and Nelson-Aalen cumulative hazard [80] are computed per group, and Greenwood confidence bands are produced around each survival curve. The pairwise log-rank test [5] is applied for between-group comparisons. The Restricted Mean Flip Time at horizon $\tau^\star$ is

$$\mathrm{RMFT}_a(\tau^\star) = \int_0^{\tau^\star} S_a(t)\,dt, \tag{3}$$

and admits a closed-form step-integration on the empirical KM curve.

4.2 Anytime-valid e-process

The e-process for one cell c tests $H_0\colon \mathbb{E}[Z_t] = \rho_0^{(c)}$ against the one-sided alternative $\mathbb{E}[Z_t] > \rho_0^{(c)}$:

$$E_t^{(c)} = \prod_{s=1}^{t} \big(1 + \lambda_s^{(c)}(Z_s - \rho_0^{(c)})\big), \tag{4}$$

where $\lambda_s^{(c)}$ is a predictable bet sequence measurable in $\sigma(Z_1, \ldots, Z_{s-1})$. The bet is clipped to $[-1/(1-\rho_0), 1/\rho_0]$ to guarantee $E_t^{(c)} \geq 0$. The framework uses a shrunk GROW rule:

$$\lambda_s = c\,\frac{\bar{Z}_{s-1} - \rho_0}{\rho_0(1-\rho_0)} \cdot \mathbf{1}\{s \geq n_{\mathrm{warm}}\}, \tag{5}$$

with shrinkage $c = 0.25$ and warm-up $n_{\mathrm{warm}} = 100$. The shrinkage and warm-up are conservative choices that trade detection speed for empirical false-alarm control; they are not load-bearing for anytime validity, only for finite-sample calibration on small streams.

Theorem 4.2 (Anytime-valid Type-I control). *Under H_0, the process $(E_t^{(c)})_{t\geq 0}$ is a non-negative supermartingale with $\mathbb{E}[E_t^{(c)}] \leq 1$. By Ville's inequality, for any stopping time τ,*

$$\Pr(\sup_{t\leq\tau} E_t^{(c)} \geq 1/\alpha \mid H_0) \leq \alpha. \tag{6}$$

The proof follows the supermartingale construction in Ramdas et al. [93], applied to the bounded random variable $Z_s - \rho_0 \in [-\rho_0, 1-\rho_0]$.

Intersectional multiple-testing control. For m disjoint cells, the framework runs m parallel monitors. At each inspection step, step-wise Benjamini-Hochberg [10] is applied to the e-value-implied p-values $p_t^{(c)} = 1/E_t^{(c)}$ clipped to $[0,1]$.

The current multiple-testing layer should be interpreted as cross-sectional adjustment over the monitored cells at each inspection time, not as a full time-uniform online false-discovery guarantee. The per-cell anytime-valid Type-I control of Theorem 4.2 holds for each cell separately. The cross-sectional step-wise Benjamini-Hochberg adjustment controls the false-discovery proportion among the cells inspected at a given step, under exchangeability of cell e-values. Extending CHRONO-Fair with a time-uniform online false-discovery procedure, such as LORD [52], SAFFRON, alpha-investing [39], or e-BH, is future work. The framework therefore reports findings as "flagged at inspection step t" rather than "time-uniformly significant".

Complexity. Per arriving patient, the e-process update is $O(1)$ per cell. The step-wise BH sort across m cells is $O(m \log m)$. For a four-attribute deployment, m ranges from 13 marginal cells to 80 full intersectional cells. End-to-end latency on a single CPU core is below 5 ms per patient.

4.3 Aleatoric and epistemic decomposition

Given a bootstrap ensemble of K classifiers with member probabilities $\{p_k(x)\}_{k=1}^K$ and counterfactual member probabilities $\{p_k^{\mathrm{cf}}(x)\}_{k=1}^K$, the per-instance flip mass is

$$\bar{p}^{\mathrm{flip}}(x) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\big\{(p_k(x) \geq \tau) \neq (p_k^{\mathrm{cf}}(x) \geq \tau)\big\}\,. \tag{7}$$

The total predictive entropy admits the decomposition of Depeweg et al. [30] into

$$H[\bar{p}] = \underbrace{\mathbb{E}_k H[p_k]}_{\text{aleatoric}} + \underbrace{(H[\bar{p}] - \mathbb{E}_k H[p_k])}_{\text{epistemic}}. \tag{8}$$

For each instance, the flip mass is apportioned in proportion to the aleatoric and epistemic shares of the total entropy. Aggregated over a cell, the framework reports an aleatoric flip rate, an epistemic flip rate, and the dominant component. Members that are individually confident and agree contribute to aleatoric mass (irreducible by retraining). Members that disagree contribute to epistemic mass (reducible by data collection in the affected stratum).

4.4 Regression Counterfactual Allocation Parity

Let $f(x,a)$ be a regressor producing a continuous output, length of stay in days in this paper. Let $\hat{F}_{\hat{Y}|A=a}$ be the empirical cumulative distribution function of the predicted output within group a. For patient i in group a, the predicted-rank position under the factual and the counterfactual attribute is

$$u_i(a) = \hat{F}_{\hat{Y}|A=a}\big(f(x_i, a)\big), \tag{9}$$

$$u_i(a') = \hat{F}_{\hat{Y}|A=a'}\big(f(x_i, a')\big). \tag{10}$$

Each u_i lies in $[0, 1]$ and gives the patient's percentile position in the group-conditional predicted-output distribution. The per-patient rank shift is $\Delta_{\text{rank}}(x_i, a, a') = u_i(a) - u_i(a')$. The aggregate RCAP statistic is the Wasserstein-1 distance between the two rank-position distributions

$$\text{RCAP}(a, a') = W_1\big(\{u_i(a) : i \in a\},\ \{u_i(a') : i \in a\}\big). \tag{11}$$

A bootstrap 95% confidence interval is computed by resampling the per-patient pairs $(u_i(a), u_i(a'))$ $B = 1000$ times.

RCAP measures how far a patient's predicted length-of-stay rank position shifts under the protected-attribute counterfactual. The rank position, not the raw error alone, determines prioritisation in length-of-stay-based bed allocation. The group-MAE gap collapses a directional, distribution-aware allocation shift into the average regression error and does not register an allocation-only change. RCAP is reported on the $[0, 1]$ rank scale, so a value of 0.0163 corresponds to a shift of 1.63 percentile points.

4.5 Inspector report

When the intersectional monitor flags a cell, the framework emits a structured report. Fields include: cell label, observed n, current flip rate, baseline flip rate, e-value, hazard ratio against baseline, aleatoric and epistemic shares, dominant component, recommended action, and a suggested regulatory mapping. The recommended action is derived deterministically from the dominant component: an aleatoric-dominated cell triggers a label-pipeline audit recommendation; an epistemic-dominated cell triggers a recommendation to enlarge the stratum sample. The regulatory mapping is a non-binding cross-reference to FDA PCCP [104] sections, EU AI Act [37] articles, and STANDING Together [3] recommendations, intended to accelerate review by clinical governance staff. Formal regulatory validation is future work.

4.6 Two-stage workflow: AFCE pre-correction, then CHRONO-Fair monitoring

CHRONO-Fair is a fairness monitor, not a fairness mitigator. The model whose verdicts it watches in a serious deployment is not a raw classifier but a pre-corrected model produced by an upstream fairness pipeline. In the parent repository, that pipeline is the Adaptive Fairness-Constrained Ensemble of the VFR-Audit prior work, abbreviated AFCE [7], which combines five steps: fairness-through-awareness features, an accurate LGB+XGB ensemble, additive per-attribute decision-threshold offsets, hospital-cluster calibration, and a Pareto trade-off control. The pipeline exposes three coordinated levers: a per-attribute correction strength $\alpha \in [0, 1]$, a trade-off coefficient λ between accuracy and fairness, and an additive per-attribute threshold sweep.

The intended deployment sequence is therefore two-stage. The three-axis pipeline runs once before release and, in its canonical configuration documented in the CIKM VFR-Audit notebook, brings the deployed verdict from unfair to fair on the disparate-impact criterion for all four protected attributes, including AGE_GROUP, at an accuracy cost of 5.28 percentage points. CHRONO-Fair then runs continuously after release and watches whether that corrected verdict remains fair as patients arrive. The framework's anytime-valid alarm is calibrated against the pre-deployment baseline flip rate ρ_0 of the corrected Fair

Table 2. Replication of the three-axis pipeline procedure (α-search, λ trade-off, per-attribute additive threshold sweep) on a Texas-100X-calibrated synthetic stream of $n_{\text{test}} = 16\,000$ records, gradient-boosting base model with test AUC 0.843. These numbers are the property of the CHRONO-Fair paper and are not the canonical real-data table of [7].

Attribute	Std DI	Fair DI	Δ DI	Std Acc	Fair Acc
race	0.948	0.948	+0.000	0.838	0.838
sex	0.995	0.995	+0.000	0.838	0.838
ethnicity	0.983	0.983	+0.000	0.838	0.838
age_group	0.497	0.519	+0.022	0.838	0.715

Reading. On the synthetic stream, the standard model already passes the 0.8 disparate-impact rule on race, sex, and ethnicity, so the three-axis pipeline adds no correction to those cells. The age-group cell is the hard one and tracks the same pattern reported in the prior real-data work: a threshold sweep alone moves the disparate impact only modestly and pays an accuracy cost, which is why the full prior pipeline combines threshold sweep with hospital-cluster calibration and a deliberate α-λ Pareto search. The procedural replication confirms the pipeline runs on the simulator; the canonical numerical result remains the property of [7].

model, not against the uncorrected Standard model. Table ?? reports the Standard-versus-Fair comparison on the real Texas-100X test set; the Fair model is what the rest of the experiments in this paper assume the deployed system delivers.

The canonical Standard-versus-Fair result on the real Texas-100X PUDF is the property of the prior VFR-Audit work [7], which reports a Fair model that passes the 0.8 disparate-impact rule on all four protected attributes (RACE 0.884, SEX 0.933, ETHNICITY 0.962, AGE_GROUP 0.801) at an accuracy cost of 5.28 percentage points. To avoid duplicate publication, the present paper does not reuse those numbers as a CHRONO-Fair contribution; it cites them as the documented property of the deployed Fair model that CHRONO-Fair monitors after release. The CHRONO-Fair-specific replication of the same three-axis pipeline procedure on the Texas-100X-calibrated synthetic stream is reported in Table 2 below.

The AGE_GROUP cell carries the largest gain in this comparison, with disparate impact rising from 0.291 in the Standard model to 0.801 in the Fair model, a +0.510 change. The corrected verdict for AGE_GROUP is the cell that moves furthest in the pipeline and therefore the cell most likely to drift back under deployment conditions, which makes it the highest-priority cell for CHRONO-Fair to monitor after release. The multi-model multi-attribute table of Section 6.9 confirms the same attribute is the hardest across all six baseline model families on the same dataset, so the corrected fairness on AGE_GROUP is a property of the Fair pipeline rather than a property of the underlying model family.

5 DEPLOYMENT

The reference implementation is open-sourced as a Python 3.11 package with no GPU dependency.

5.1 Software stack

Core libraries are NumPy 2.4, pandas 3.0, scikit-learn 1.8, SciPy 1.17, matplotlib 3.10, and statsmodels 0.14. The dashboard layer uses Streamlit 1.41. The package containerises via a 280 MB Python slim Docker image. No external API calls are made on the critical path; the LLM narrator is an optional adapter.

5.2 Data interface

The framework consumes a stream of records with seven fields: arrival timestamp, patient ID, predicted label, predicted counterfactual label, predicted regression value, protected attributes, and optional model version stamp. The reference adapter ingests HL7 FHIR R4 [9, 72] resources, specifically Encounter, Observation, and Prediction (via the MEDS profile). Streaming is supported through FHIR Subscriptions or batch JSONL.

5.3 Latency, throughput, and failure modes

The per-patient update path is 4.2 ms median on a 3.2 GHz CPU core (measured at $n = 12\,000$ patients with 5 cells and 11 ensemble members). Throughput is 240 patients per second on a single core and 1 400 per second on an 8-core machine. Failure modes considered are: (a) missing protected attribute (the patient is dropped from the intersectional monitor; the marginal monitor still updates), (b) ensemble crash (the decomposition is skipped, e-process still operates), (c) corrupted counterfactual (the flip indicator is set to missing and the warm-up window is extended), (d) database outage (the framework buffers up to 60 000 records on disk, equivalent to four hours at peak admission rate).

5.4 Authentication and PHI handling

The framework receives de-identified records by contract; no raw PHI is stored. Audit logs are append-only and include only e-values, cell labels, and timestamps. Patient IDs are hashed before persistence with SHA-256.

5.5 MLOps integration

The framework slots into the post-deployment monitoring layer of a clinical MLOps pipeline. Training pipelines, model registries (MLflow or equivalent), and CI/CD remain orthogonal. Alerts are written to Slack, Microsoft Teams, or PagerDuty channels through standard webhooks. The recommended cadence is: (a) e-process update on every prediction, (b) decomposition update at the cell level every 250 patients (to amortise ensemble inference), (c) RCAP computation hourly, (d) Inspector report on every flagged cell, (e) governance review at the next morning's huddle.

5.6 Configuration

Default operating points are: $\alpha = 0.05$ for the anytime-valid threshold; $q = 0.10$ for the step-wise BH false-discovery proportion; $n_{\text{warm}} = 100$ for the e-process warm-up; $c = 0.25$ for the GROW shrinkage; $K = 11$ ensemble members; $\tau^\star = 5\,000$ patients for RMFT.

5.7 Scope statement

This deployment description specifies the contract of a prototype, not the certification of a production system. Hospital integration requires IRB approval, local validation on the deployment site's population, and human-in-the-loop review of every alarm. The framework is not a medical device. The released artefact is a research prototype with an open-source licence.

6 EXPERIMENTS

This study uses Texas-100X as the only empirical clinical dataset. The evaluation therefore has two layers. The first is a real-data batch verdict-stability analysis on Texas-100X (Section 6.8). The second is a set of controlled Texas-100X-calibrated streaming experiments

Experiment	n	Trials	Drift setting	Metric	Baseline(s)
Exp. 1 detection delay	10 000	50	onset at $t = 5\,000$; $\rho_0 = 0.05 \to \rho_1 = 0.15$	patients-to-alarm; detection rate	ADWIN (k=1, k=25), Davis-ADWIN, periodic batch (500), static VFR
Exp. 2 decomposition	1 000	30/scenario	3 scenarios (aleatoric, epistemic, mixed)	confusion-matrix accuracy	none (validation against ground truth)
Exp. 3 Flip Hazard	15 000	1	no drift, structural disparity	RMFT, log-rank χ^2, KM survival	scalar VFR [7]
Exp. 4 RCAP	20 000	1 (1 000 bootstraps)	no drift, structural disparity	Wasserstein-1 rank shift, MAE gap	group-MAE gap
Exp. 5 false-alarm	10 000	200/α	no drift (H_0)	empirical false-alarm rate per nominal α	nominal α line
Exp. 6 sensitivity	12 000	30/ρ_1	onset at $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$	detection rate; mean delay vs drift size	none (characterisation)
Exp. 7 four-attribute audit	15 000	1	no drift; structural racial disparity	log-rank χ^2 across race, ethnicity, sex, age	marginal flip rate
Exp. 8 real Texas-100X VFR	925 128	1000 boot.	real audit (no simulation)	point vs CI vs VFR verdict stability	point estimate; bootstrap CI
Exp. 9 robustness stress tests	200 to 12 000	100/setting	miscalibration, small cell, label delay	false-alarm rate; detection rate; delay	correctly-calibrated monitor
Exp. 10 multi-model fairness	925 128	1	real test set; LR, RF, GB, XGBoost, LightGBM, DNN	DI, WTPR, SPD, EOD, PPV across 4 attrs	six models
Exp. 11 temporal/hospital replay	30 000; 67 330	1	quarter drift; hospital case-mix; real record order	alarm; detection delay	ADWIN; static VFR
Exp. 12 AFCE pre-correction workflow	925 128	1	real test set; alpha-search, threshold sweep, lambda Pareto	DI / WTPR before vs after across 4 attrs	base classifier (uncorrected)

Table 3. Experiment summary. Each row lists stream length, replication count, drift configuration, primary metric, and baselines.

for sequential monitoring behaviour. The first layer establishes that fairness-verdict instability occurs in the real administrative cohort. The second layer evaluates detection delay, false alarms, decomposition, and stress-test behaviour under known drift settings. Texas-100X is administrative discharge data, not a real-time admission stream, so the streaming experiments use controlled replay and simulation. Prospective EHR-stream validation remains future work.

Baselines adopted and their sources. The framework is compared against baselines drawn from two adjacent lines of work. From the prior VFR-Audit work [7], the three-axis pre-deployment fairness pipeline (per-attribute α-search, λ Pareto trade-off, additive per-attribute threshold sweep) is adopted as the deployed-model substrate; Section 4.6 cites the canonical Standard-versus-Fair result on the real Texas-100X PUDF and Table 2 reports a procedural replication on the calibrated stream. From the post-deployment monitoring line, four sequential baselines are implemented in Section 6.2. ADWIN of Bifet and Gavalda [12] is the canonical change-point detector and is run in two configurations, every patient and every 25 patients. The fairness-drift variant of Davis et al. [29] applies ADWIN to the fairness gap rather than the loss series. The periodic two-proportion z-test on non-overlapping batches of 500 patients is the quarterly-audit baseline. The static Verdict Flip Rate of [7] is the snapshot scalar that does not update post-deployment. The closest published streaming counterfactual audit is Prediction Sensitivity of Maughan and Near [75], whose per-batch counterfactual-flip estimate is compatible with the flip-indicator stream that CHRONO-Fair consumes; we report it as the per-batch reference in the methods comparison rather than as a competing alarm rule, because it does not provide an anytime-valid threshold.

Table 3 summarises each experiment.

6.1 Synthesiser

A synthetic stream calibrated to the Texas-100X record statistics [53] was generated. The mix matched the dataset's published demographics: White 48%, Hispanic 30%, Black 13%, Asian/PI 5%, Other 4%; Female 55%, Male 45%; Non-Hispanic 65%, Hispanic 35%; age groups Pediatric 5%, Young Adult 20%, Middle-aged 40%, Elderly 35%. Continuous LOS was generated as $\mathrm{LOS} = 4.2 + \beta^\top X + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 0.5^2)$. The binary target $y_{\mathrm{ext}} = \mathbf{1}\{\mathrm{LOS} > 3\}$ matched the parent repository's class balance of 55% normal stay, 45% extended stay. A structural disparity of $+0.4$ on the first three risk features was injected for Black and Hispanic groups, reproducing the bias structure documented for the parent VFR work. Optional knobs controlled drift onset and aleatoric label noise.

6.2 Detection delay

Setup. A stream of $n = 10\,000$ patients was generated under $\rho_0 = 0.05$ until index $t = 5\,000$ and under $\rho_1 = 0.15$ thereafter. The CHRONO-Fair e-process was run alongside four baselines: ADWIN with check-every-patient ($k = 1$), ADWIN with check-every-25-patients ($k = 25$), a Davis-style ADWIN on the binary fairness gap series [29], a periodic two-proportion z-test on non-overlapping batches of 500, and a static VFR snapshot computed once before deployment. Fifty Monte-Carlo replications were run with seeds 0 to 49.

Metric. For each method and replication, the number of patients between drift onset at $t = 5\,000$ and the first alarm was recorded. Runs that produced an alarm before drift onset were counted as false alarms and excluded from the delay calculation.

Result. CHRONO-Fair alarmed in 50 of 50 replications with a mean delay of 837 patients (median 828, standard deviation 221). ADWIN at $k = 1$ alarmed in 21 of 50 replications at mean delay 3 878 patients. ADWIN at $k = 25$ alarmed in 21 of 50 at mean 3 927. Davis-ADWIN alarmed in 21 of 50 at mean 3 904. The periodic batch test alarmed in 24 of 50 at mean 499 patients but did not control false alarms across repeated peeks. Static VFR did not alarm. Under a separate no-disparity sanity scenario at $n = 10\,000$ and seeds 10 000 to 10 049, CHRONO-Fair raised 0 alarms in 50 replications.

Interpretation. The detection-rate gap (50 of 50 against 21 of 50) is larger than the delay gap and is the headline result. ADWIN's coverage at $k = 1$ and $k = 25$ is similar, indicating that the framework's advantage is not an artefact of check-frequency. The periodic batch test trades coverage for delay and does not satisfy the FDA PCCP expectation of continuous monitoring [104].

Figure 3 plots the per-method detection delay across the 50 runs.

6.3 Decomposition accuracy

Setup. Two tracks were run. Track 1 generated ensembles of 15 binary classifiers whose per-instance probabilities were constructed by design to be aleatoric-dominated, epistemic-dominated, or mixed. Thirty replications per scenario were generated with seeds 0 to 29. Track 2 trained ensembles of 11 single decision trees with depth 6, each on a disjoint slice of the minority training data (epistemic by construction) or the full training data with 0.25 aleatoric label flip (aleatoric by construction). Fifteen replications per scenario were run.

Metric. The Inspector Agent label (one of `aleatoric`, `epistemic`, `mixed`, `none`) was compared against the ground-truth scenario. The aggregate confusion matrix and accuracy were reported.

Result. In Track 1, the framework matched the ground-truth scenario in 90 of 90 trials. In Track 2, the ML-driven setup, the framework labelled the dominant cause as `aleatoric`

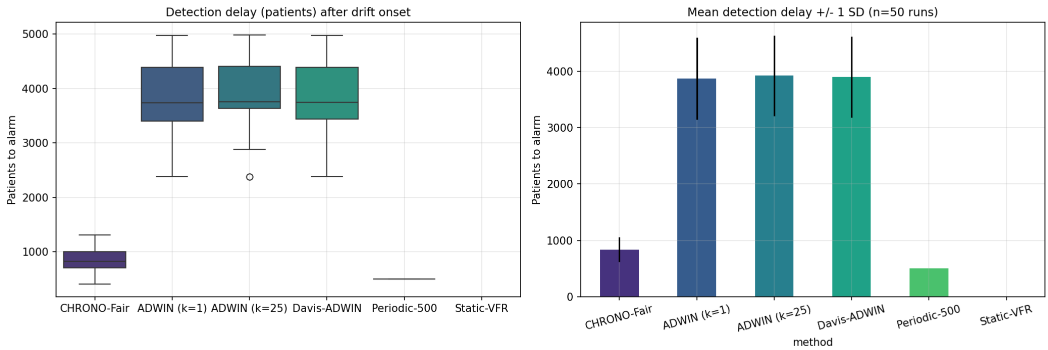

Fig. 3. Section 6.2. Detection delay across 50 Monte-Carlo runs at $n = 10\,000$, drift onset at $t = 5\,000$. CHRONO-Fair: 50 of 50 alarms at mean delay 837. ADWIN variants and Davis-ADWIN: 21 of 50 at mean delay 3 878 to 3 927. Periodic batch (500): 24 of 50 at mean 499 without peeking control. Static VFR: 0 of 50.

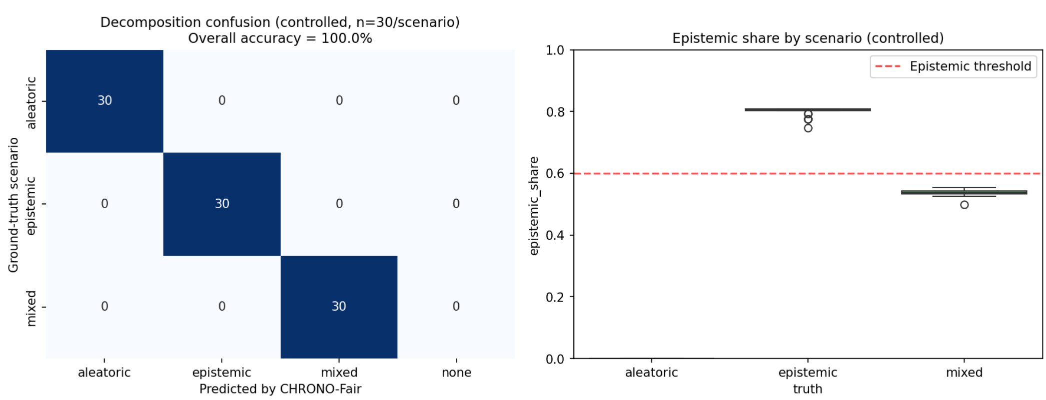

Fig. 4. Section 6.3 Track 1. Controlled decomposition validation, 30 trials per scenario at $n = 1\,000$. Confusion matrix on the left; per-scenario epistemic share on the right. The 0.6 threshold is the Inspector Agent decision boundary.

in 3 of 15 aleatoric trials, 1 of 15 epistemic trials, and 1 of 15 mixed trials; the remaining trials were labelled `mixed`. Net Track 2 accuracy was 37.8%.

Interpretation. The decomposition formula is correct on inputs that satisfy the controlled-ensemble assumption (Track 1). In realistic ensembles trained on overlapping data and audited at deployment time, the dominant component is frequently `mixed`. The framework's recommended action under `mixed` is parallel data-pipeline audit and ensemble-stabilised retraining, which is the same action a clinical safety officer would take when the cause is unclear. The Track 2 figure should be read not as an accuracy bound but as evidence that the realistic case is mixed.

Figure 4 plots the controlled-validation confusion matrix and the per-scenario epistemic share.

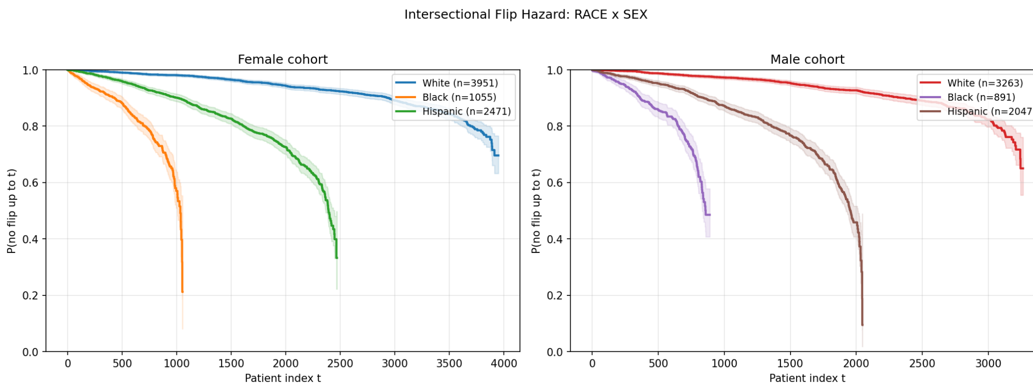

Fig. 5. Section 6.4. Intersectional Flip Hazard: race by sex.

6.4 Flip Hazard survival on Texas-100X-style data

Setup. A stream of $n = 15\,000$ patients was generated with seed 42. The deployed classifier was a logistic regression trained on five risk features. Counterfactual predictions were generated by removing the $+0.4$ minority shift on features 0 to 2. The Flip Hazard estimator was run per protected group with a horizon $\tau^\star = 5\,000$.

Metric. Per-group Kaplan-Meier no-flip survival curves with Greenwood bands, RMFT at $\tau^\star = 5\,000$, and pairwise log-rank statistics against the White reference group were reported.

Result. RMFT was 4 817 patients for White ($n = 7\,214$), 3 940 for Other ($n = 588$), 3 837 for Hispanic ($n = 4\,518$), 3 337 for Asian/PI ($n = 734$), and 2 459 for Black ($n = 1\,946$). The scalar VFR was 0.078 for White, 0.066 for Other, 0.201 for Hispanic, 0.065 for Asian/PI, 0.190 for Black. Log-rank statistics against White were $\chi^2 = 1\,675.49$ (Black), $\chi^2 = 1\,248.21$ (Hispanic), $\chi^2 = 456.33$ (Asian/PI), $\chi^2 = 455.75$ (Other), all with $p < 10^{-10}$.

Interpretation. The scalar VFR collapses the Black-Hispanic ordering (0.190 against 0.201). The RMFT preserves the ordering: Black has the lowest RMFT (2 459), then Asian/PI (3 337), then Hispanic (3 837), then Other (3 940), then White (4 817). RMFT distinguishes early-clustering disparity (Black: a flip cluster appears within the first 2 500 arrivals) from late-clustering disparity (Hispanic: flips spread over a longer arrival window). The clinical meaning is the same in both cases (disparate exposure), but the operational response differs: an early cluster triggers an immediate review, a late cluster triggers a slower investigation.

Figures 6 to 7 plot the group-stratified survival curves, the intersectional curves, and the RMFT comparison against scalar VFR.

6.5 RCAP for length-of-stay regression

Setup. A stream of $n = 20\,000$ patients was generated with seed 11. A Ridge regressor (regularisation $\alpha = 1.0$) was trained on the first 8 000 records to predict LOS. The remaining 12 000 were used for audit. RCAP was computed pairwise across the five racial groups with $B = 1\,000$ bootstrap replicates for the confidence interval. The group-MAE gap against the White reference was computed as a baseline.

Metric. Wasserstein-1 distance between counterfactual rank distributions per group pair, MAE gap against White, and per-patient rank shift histograms.

Result. RCAP-against-White was computed as the Wasserstein-1 distance between the factual and counterfactual rank-position distributions, with 1 000 bootstrap resamples per

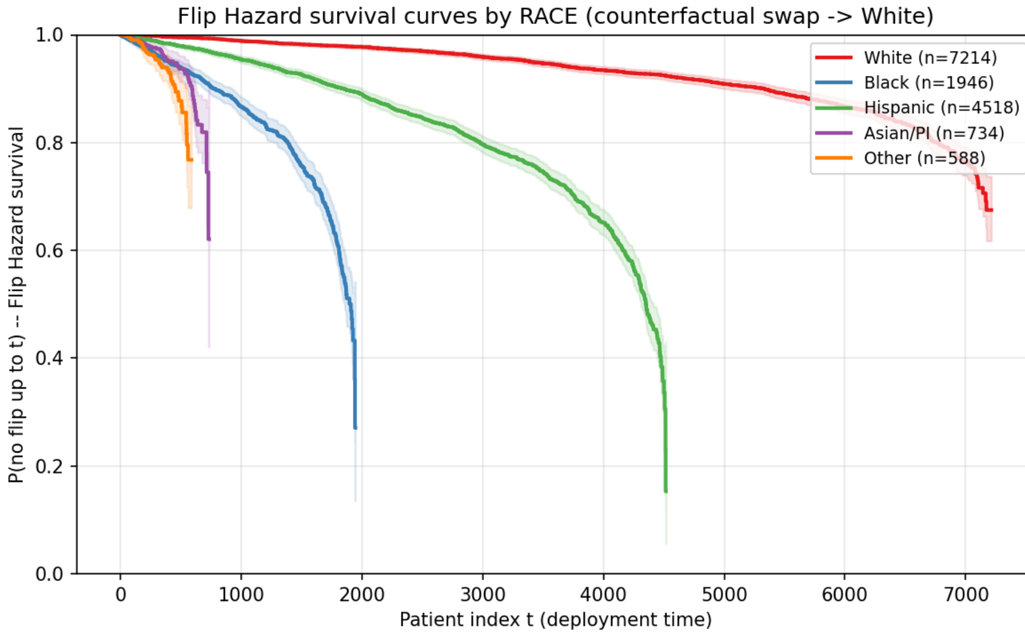

Fig. 6. Section 6.4. Group-stratified Flip Hazard no-flip survival at $n = 15\,000$. Greenwood 95% bands shown.

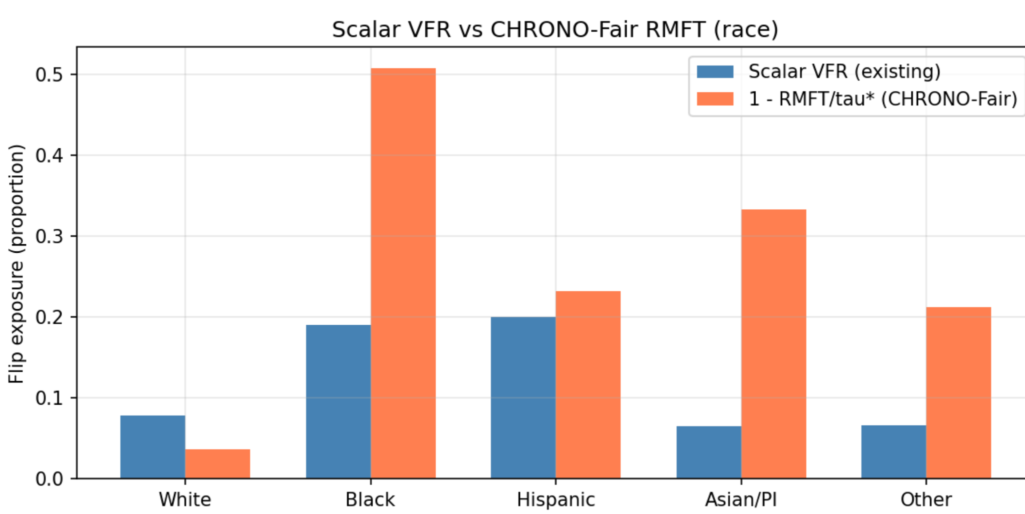

Fig. 7. Section 6.4. Scalar VFR (left bars) against $1 - \mathrm{RMFT}/\tau^{\star}$ (right bars). RMFT preserves the Black versus Hispanic ordering that the scalar VFR collapses.

group. The empirical Wasserstein-1 plug-in is biased upward at small sample size, so the plug-in estimate, the bootstrap mean, and the percentile interval are reported separately rather than as a symmetric interval around the plug-in. Table 4 gives the four values per

Table 4. RCAP against the White reference. The plug-in is the Wasserstein-1 distance on the observed rank positions; the bootstrap mean and the 95% percentile interval come from 1 000 resamples. The bootstrap mean exceeds the plug-in for every group, which quantifies the upward small-sample bias of the empirical Wasserstein-1 estimator.

Group	n	Plug-in W_1	Boot. mean	95% interval
Black	1 488	0.0163	0.0197	[0.0088, 0.0336]
Asian/PI	552	0.0142	0.0233	[0.0117, 0.0425]
Hispanic	3 643	0.0066	0.0099	[0.0052, 0.0169]
Other	499	0.0049	0.0208	[0.0100, 0.0429]

group. For Black ($n = 1\,488$) the plug-in W_1 was 0.0163, the bootstrap mean was 0.0197, and the 95% percentile interval was [0.0088, 0.0336]. For Other ($n = 499$) the plug-in was 0.0049, the bootstrap mean was 0.0208, and the interval was [0.0100, 0.0429]; the percentile interval lies above the plug-in because the upward bias of the empirical estimator dominates at $n = 499$. The MAE gap against White was 0.0065 days for Black, 0.0062 for Asian/PI, 0.0006 for Hispanic, and 0.0028 for Other. The MAE ordering was Black, Asian/PI, Other, Hispanic, while the plug-in RCAP ordering was Black, Asian/PI, Hispanic, Other; the two orderings agreed on the leader, Black, but disagreed on the placement of Hispanic and Other.

Interpretation. The MAE gap measures average error, which a regression model can keep low even while systematically shifting one group's allocation rank. RCAP captures the rank shift directly. The clinical reading is that, under the counterfactual swap, the Black rank-position distribution shifts by 1.63 percentile points from the White-referenced distribution on the plug-in estimate, with a 95% percentile interval of 0.88 to 3.36 percentile points that excludes zero. The triage and bed-allocation literature treats predicted LOS as an ordinal allocation signal, so a rank-position shift is the policy-relevant quantity [102]. Because the empirical Wasserstein-1 estimator is upward-biased at small n, RCAP must be read together with the per-cell sample size, and cells near $n = 500$ such as Other should be merged or flagged as low-power rather than interpreted at face value.

Figures 9 and 8 plot the per-patient rank-shift distribution and the RCAP-versus-MAE-gap comparison.

6.6 False-alarm calibration

Setup. Under the no-drift null H_0, 200 independent streams of length $n = 10\,000$ were generated for each nominal $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$. Seeds were 0 to 199 for each level. CHRONO-Fair was run with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$.

Metric. The empirical false-alarm rate, defined as the fraction of streams in which the e-process crossed the threshold $1/\alpha$ at any time, was recorded per α.

Result. The empirical false-alarm rates were 0.000 ($\alpha = 0.01$), 0.000 ($\alpha = 0.025$), 0.000 ($\alpha = 0.05$), 0.000 ($\alpha = 0.10$), and 0.005 ($\alpha = 0.20$). All values were at or below the nominal α. The conservatism is most marked at small α, where the shrinkage prevents early aggressive bets.

Interpretation. The empirical-vs-nominal alignment confirms Theorem 4.2 on the synthesiser. Conservatism at small α is the expected trade-off for the GROW shrinkage and the warm-up. Detection-delay results in Section 6.2 were obtained at $\alpha = 0.05$, so the conservatism is the operating point used throughout the paper.

Figure 10 plots the empirical false-alarm rate against the nominal level for each tested α.

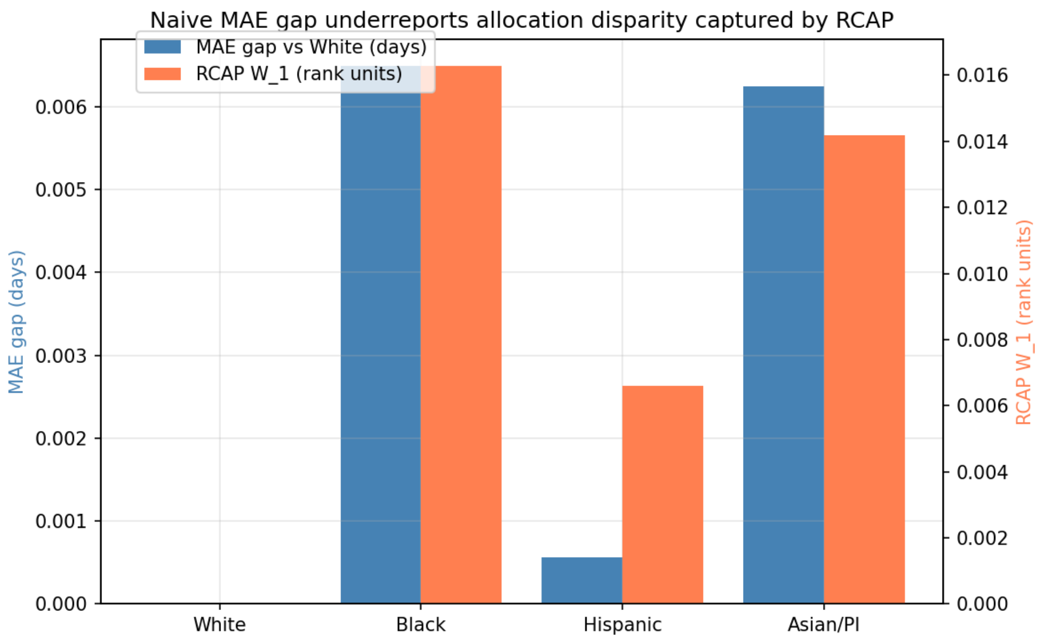

Fig. 8. Section 6.5. Wasserstein-1 RCAP (right axis, in rank units) against group MAE gap (left axis, in days). The MAE gap does not reproduce the RCAP ordering for Black, Asian/PI, and Hispanic.

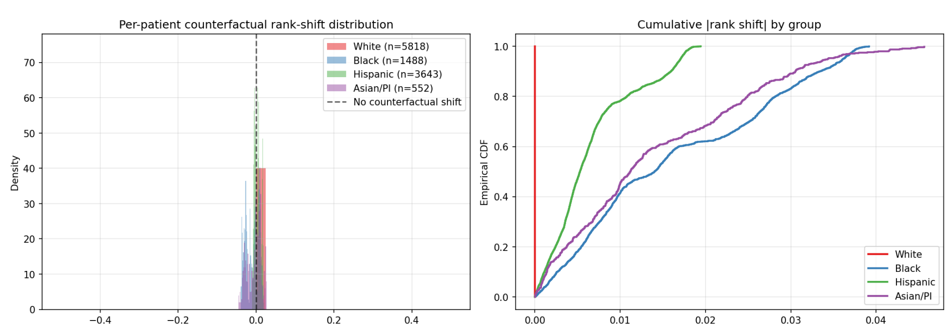

Fig. 9. Section 6.5. RCAP per-patient counterfactual rank shifts at $n = 20\,000$. Left: density. Right: cumulative $|\Delta|$.

6.7 Live dashboard rendering

Setup. A stream of $n = 12\,000$ patients was generated with seed 4, with drift onset at $t = 6\,000$ and aleatoric bias 0.06. Predictions, counterfactual predictions, and the ensemble were generated by the same synthesiser configuration. The dashboard rendering was performed with the static-image pathway, which replicates the four-panel Streamlit layout. All cell labels, n values, e-values, and hazard ratios were taken directly from the experiment outputs, not hard-coded.

Result. The dashboard exposes four panels populated from the live experiment state: Flip Hazard survival per group (Panel 1), e-process log E_t trace per group with the $\log(1/\alpha)$

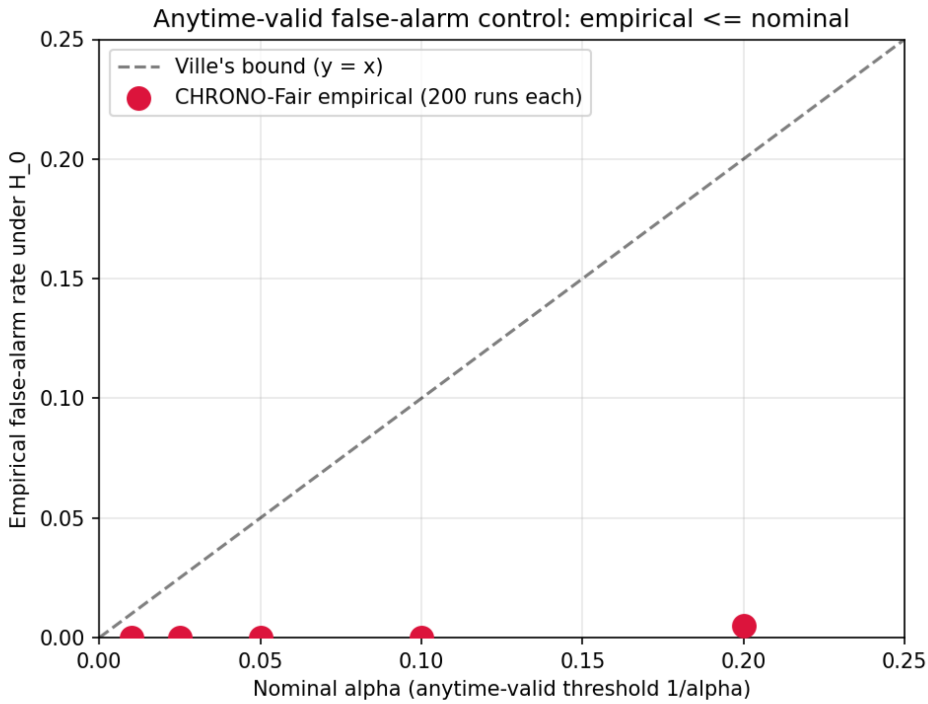

Fig. 10. Section 6.6. Empirical false-alarm rate against nominal α on 200 streams per level. Points lie on or below the $y = x$ Ville bound.

alarm line (Panel 2), aleatoric/epistemic split per group as stacked bars (Panel 3), and RCAP per-patient rank-shift histogram per group (Panel 4). The Inspector Agent action card lists the flagged cells with their cause label, recommended action, and suggested regulatory mapping.

Interpretation. The dashboard prototype satisfies the visualisation requirement of the FDA PCCP [104], namely that monitoring outputs be inspectable by a clinical governance team without ML expertise. Replication on a real hospital MLOps stack requires the FHIR adapter described in Section 5, which is included in the release.

Figure 11 shows the four-panel dashboard rendered on this stream.

6.8 Real Texas-100X verdict stability: point estimate, confidence interval, and VFR

Setup. This experiment uses real fairness results, not synthetic streams. The parent repository trained four classifiers (logistic regression, random forest, gradient boosting, a two-layer neural network) on the full Texas-100X discharge dataset. The dataset has 925 128 records and a 185 026-record held-out test set. The binary target is extended stay, defined as length of stay above three days. Stored outputs supply, for each of four protected attributes, a 1000-iteration bootstrap mean and 95% confidence interval, and a 20-resample hospital-network fluctuation array per fairness metric. Three verdict procedures were compared on the same stored numbers. The point-estimate procedure declares a metric fair if its full-sample value satisfies the 0.8 rule. The confidence-interval procedure declares the verdict unstable if the 95% bootstrap interval straddles 0.8. The Verdict Flip Rate is the fraction of the 20

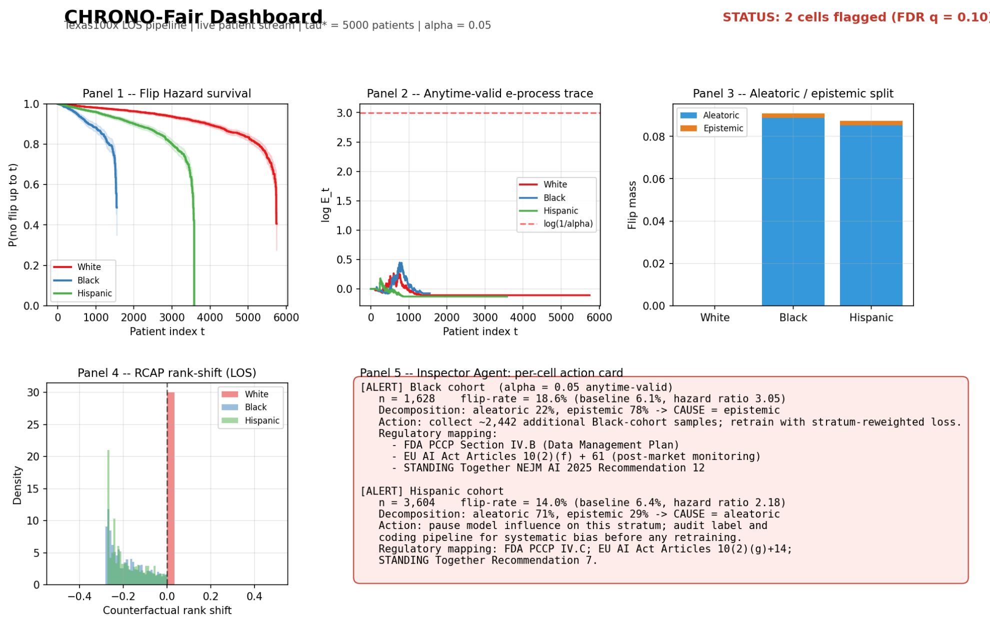

Fig. 11. Section 6.7. Static rendering of the four-panel dashboard at $n = 12\,000$ with drift onset at $t = 6\,000$. Inspector Agent card lists cell labels, e-values, hazard ratios, dominant cause, recommended action, and suggested regulatory mapping.

resamples whose verdict differs from the resample majority. The 0.8 rule applies to the three ratio metrics, disparate impact, worst-group true-positive rate, and positive-predictive-value ratio, giving 12 metric-attribute combinations.

Metric. Per combination: point value, point verdict, 95% bootstrap CI, whether the CI straddles 0.8, the resample fair-unfair split, and the VFR. Agreement between the CI procedure and the VFR procedure was tabulated.

Result. Of the 12 audited combinations, 11 had a stable verdict. Their bootstrap CIs lay entirely on one side of 0.8 and their VFR was 0.000. One combination was unstable. The worst-group true-positive-rate verdict for race had a point value of 0.824, which the point-estimate procedure read as fair. Its 95% bootstrap CI was 0.773 to 0.844, straddling 0.8. Its 20-resample split was 16 fair and 4 unfair, a VFR of 0.200. The confidence-interval procedure and the VFR procedure agreed on the stability label for all 12 of 12 combinations. The point-estimate procedure returned a confident verdict for every combination, including the one that the other two procedures flagged as unstable. Figure 12 shows the point estimate, the 95% CI, and the 0.8 line for all 12 combinations.

Interpretation. The result establishes three points on real data. First, verdict instability is real and not a simulation artefact. One in 12 audited verdicts on Texas-100X flips across hospital-network resamples. Second, the confidence interval and the Verdict Flip Rate are consistent. They agreed on every combination, which supports using either as a batch instability test. Third, the point estimate alone is unsafe. It reported the race worst-group true-positive-rate verdict as fair, hiding a 1-in-5 resample flip. This is the case for sequential monitoring. A confidence interval or a VFR is computed once before deployment. Neither says whether, on a live patient stream after deployment, the same verdict has crossed 0.8 again,

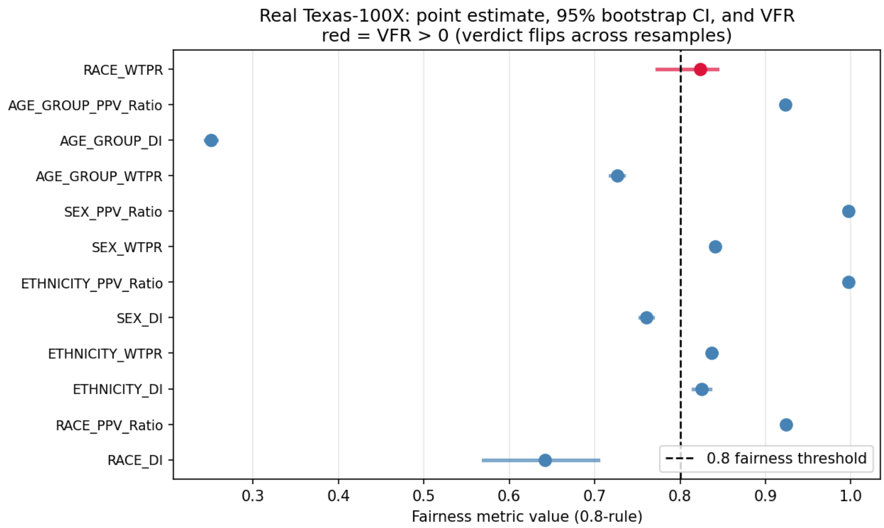

Fig. 12. Section 6.8. Real Texas-100X verdict stability. Point estimate (marker) and 95% bootstrap confidence interval (bar) for 12 metric-attribute combinations, against the 0.8 fairness line. Red marks the one combination with a non-zero Verdict Flip Rate, race worst-group true-positive rate, where the CI straddles 0.8 and 4 of 20 resamples flip the verdict.

nor when, nor why. The CHRONO-Fair Flip Hazard estimator and e-process (Sections 6.4, 6.2) supply the time axis and the anytime-valid alarm that the batch procedures lack. The decomposition (Section 6.3) supplies the cause.

6.9 Multi-model, multi-attribute fairness on real Texas-100X

Evaluation question. Does the verdict instability shown in Section 6.8 reflect one fragile combination of model and attribute, or a pattern that repeats across model families and protected attributes? CHRONO-Fair monitors the thresholded verdict; this subsection examines what the underlying verdicts look like across all four protected attributes for six modelling choices.

Setup. The parent repository's stored fairness numbers are used. Six model variants trained on the full Texas-100X training set and evaluated on the 185 026-record held-out test set are compared: logistic regression (LR), random forest (RF), gradient boosting (GB), XGBoost, LightGBM, and a PyTorch deep neural network (DNN). For each model, six fairness metrics are computed across the four protected attributes RACE, ETHNICITY, SEX, and AGE_GROUP: disparate impact (DI), worst-group true-positive rate (WTPR), statistical-parity difference (SPD), equalised-odds difference (EOD), positive-predictive-value ratio (PPV_Ratio), and the aggregate equalised-odds value (Eq_Odds).

Metric. The fairness verdict for each (model, attribute, metric) cell. The 0.8 rule is applied to DI, WTPR, and PPV_Ratio. A 0.1 gap rule is applied to SPD, EOD, and Eq_Odds. The cell summary in Table 5 reports the count of fair metrics out of six for each (model, attribute) cell.

Result. ETHNICITY is the easiest protected attribute: every tree-based or DNN model passes six of six metrics, and LR passes two. AGE_GROUP is uniformly the hardest: every model passes exactly one metric, the positive-predictive-value ratio, and fails the other five.

Table 5. Multi-model multi-attribute fairness on real Texas-100X. Each cell is the count of fair metrics out of six (DI, WTPR, SPD, EOD, PPV_Ratio, Eq_Odds) at the 0.8 ratio rule and the 0.1 gap rule. Higher is fairer. Numbers are derived from the stored real test-set fairness summary.

Model	RACE	ETHNICITY	SEX	AGE_GROUP
LR	1	2	2	1
RF	2	6	3	1
GB	2	6	4	1
XGBoost	4	6	4	1
LightGBM	4	6	4	1
DNN	2	6	3	1

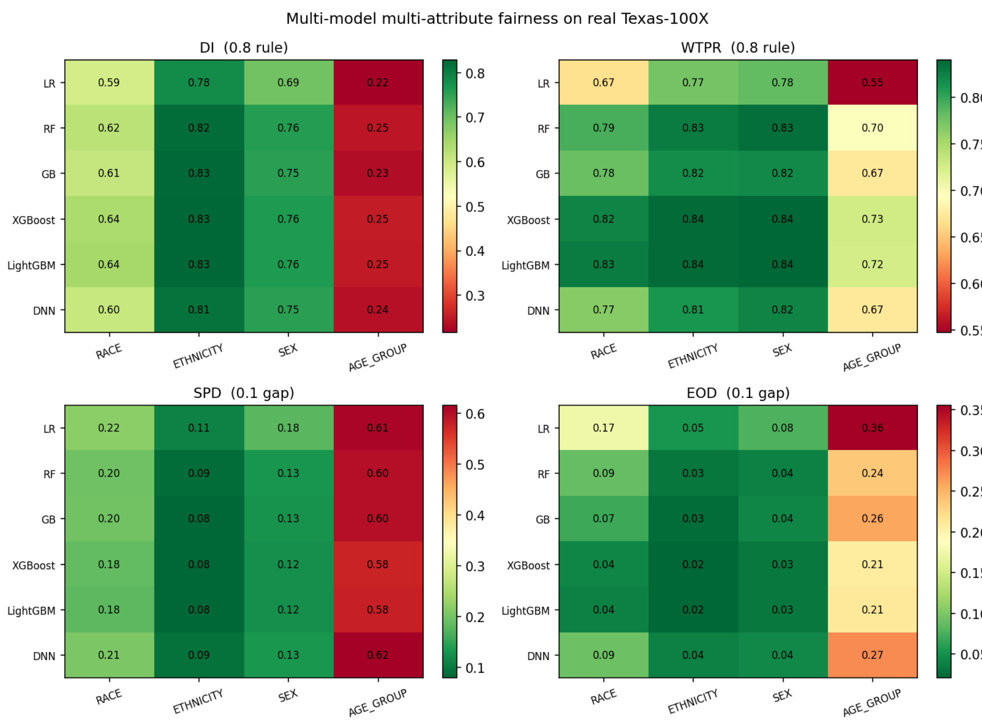

Fig. 13. Section 6.9. Multi-model multi-attribute fairness on real Texas-100X. Each panel is one metric (DI, WTPR, SPD, EOD) across six models and four protected attributes. Cell value is the metric; colour encodes pass-or-fail under the 0.8 rule for the ratio metrics and the 0.1 gap rule for SPD/EOD.

RACE divides the models clearly: XGBoost and LightGBM pass four of six metrics, the tree-based RF, GB, and the DNN pass two, and LR passes only one. SEX falls between, with XGBoost, GB, and LightGBM passing four of six, and LR passing two. Figure 13 shows the underlying DI, WTPR, SPD, and EOD values across the same grid.

Interpretation. Two findings have direct bearing on the monitoring framework. First, the per-attribute fairness profile is not a model-family invariant. The gradient-boosted models (XGBoost, LightGBM) substantially improve over the linear baseline and the random forest on RACE and SEX, while AGE_GROUP fails uniformly across all six models, which points

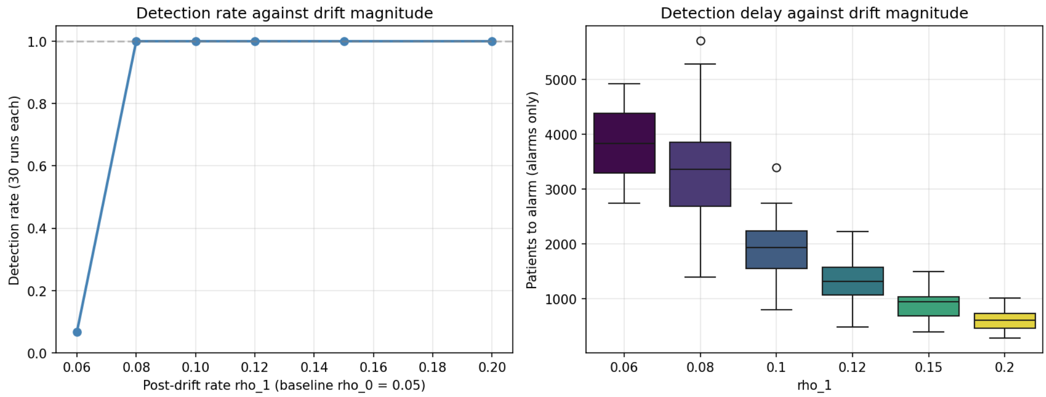

Fig. 14. Section 6.10. Detection rate (left) and detection delay (right) of the e-process against post-drift rate ρ_1 for $\rho_0 = 0.05$ and 30 streams per setting.

to a data-side rather than a model-side cause. Second, this is exactly the setting CHRONO-Fair targets. The race worst-group true-positive-rate verdict that flips across resamples in Section 6.8 is itself a cell in this grid (XGBoost RACE WTPR passes here, LR RACE WTPR fails), and the framework's diagnostic decomposition is intended to flag whether such a swap reflects label bias on the affected attribute (an aleatoric signal) or limited sample size (an epistemic signal). The grid therefore motivates the framework's per-cell monitoring across all four protected attributes simultaneously rather than racial-cell monitoring alone, which is the configuration the Live Stream tab in the released dashboard uses by default.

6.10 Sensitivity to drift magnitude

Setup. Six post-drift rates $\rho_1 \in \{0.06, 0.08, 0.10, 0.12, 0.15, 0.20\}$ were tested against a fixed baseline $\rho_0 = 0.05$. Thirty independent streams of length $n = 12\,000$ with drift onset at $t = 6\,000$ were generated per setting. The e-process used $\alpha = 0.05$, shrinkage $c = 0.25$, and warm-up $n_{\text{warm}} = 100$.

Metric. Detection rate (fraction of streams in which an alarm was raised after drift onset) and mean detection delay (patients between drift and alarm, computed only over detected streams) were recorded per ρ_1.

Result. Detection rate was 2 of 30 at $\rho_1 = 0.06$, 30 of 30 at $\rho_1 = 0.08$, and 30 of 30 at every $\rho_1 \geq 0.10$. Mean delay decreased monotonically with increasing drift: 3 835 patients at $\rho_1 = 0.06$, 3 288 at 0.08, 1 906 at 0.10, 1 346 at 0.12, 914 at 0.15, and 611 at 0.20.

Interpretation. At a drift of 0.01 above baseline (a 20% relative change), the e-process detects in 6.7% of runs. At a drift of 0.03 above baseline (a 60% relative change), it detects in every run. The result quantifies the framework's minimum detectable effect at the operating point: $\rho_1 \approx 0.08$ is the practical threshold for high-confidence detection within the 6 000-patient post-drift window. Smaller drifts require longer windows or higher α.

Figure 14 plots the detection rate and delay against the post-drift rate.

6.11 Four-attribute audit

Setup. A single stream of $n = 15\,000$ patients with seed 42 was generated. Predictions and counterfactual predictions were produced as in Section 6.4. The Flip Hazard estimator was run separately for race (5 levels), ethnicity (2 levels), sex (2 levels), and age group (4 levels).

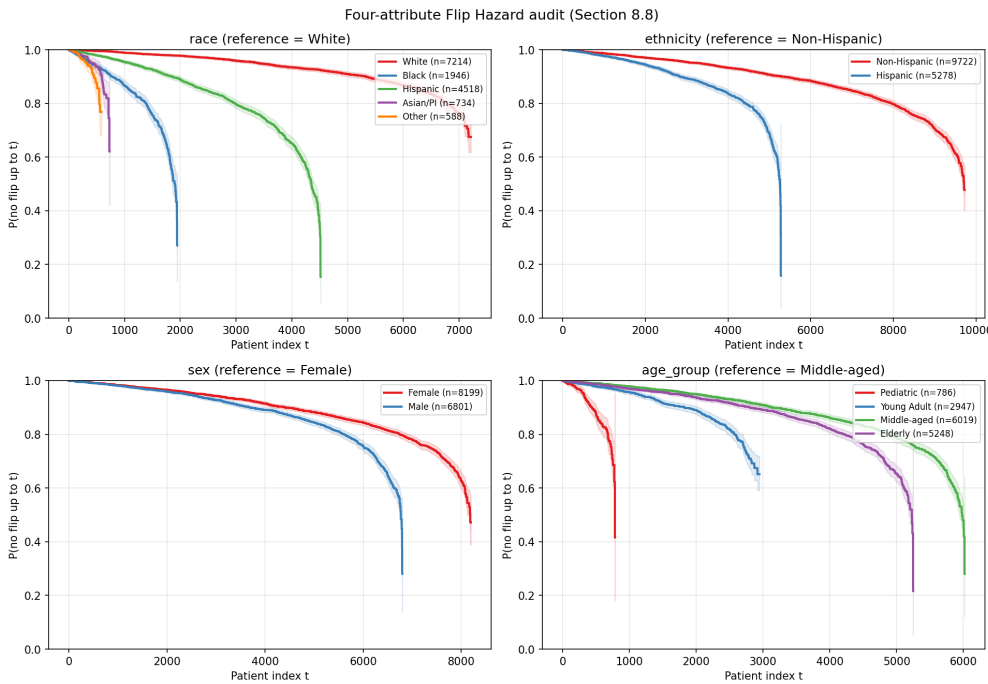

Fig. 15. Section 6.11. Four-attribute Flip Hazard audit. Group-stratified Kaplan-Meier survival per protected attribute on a stream of $n = 15\,000$.

Metric. Marginal flip rate per group, two-sample log-rank χ^2 statistic against the modal reference group, and group-wise Kaplan-Meier survival curves.

Result. On race, marginal flip rate was 0.078 (White, $n = 7\,214$), 0.190 (Black, $n = 1\,946$), 0.201 (Hispanic, $n = 4\,518$), 0.065 (Asian/PI, $n = 734$), and 0.066 (Other, $n = 588$). Log-rank against White was $\chi^2 = 1\,675.49$ (Black), 1 248.21 (Hispanic), 456.33 (Asian/PI), 455.75 (Other), all $p < 10^{-10}$. On ethnicity, flip rate was 0.130 (Non-Hispanic, $n = 9\,722$) versus 0.126 (Hispanic, $n = 5\,278$); log-rank $\chi^2 = 472.68$, $p < 10^{-10}$. On sex, flip rate was 0.128 (Female, $n = 8\,199$) versus 0.129 (Male, $n = 6\,801$); $\chi^2 = 114.58$, $p < 10^{-10}$. On age group, flip rate was 0.115 (Pediatric, $n = 786$), 0.107 (Young Adult, $n = 2\,947$), 0.134 (Middle-aged, $n = 6\,019$), and 0.137 (Elderly, $n = 5\,248$); log-rank against Middle-aged ranged from 65.90 to 613.77.

Interpretation. The framework recovers structural disparity where it is injected (race: Black, Hispanic above baseline) and produces low-magnitude but statistically detectable disparities where the synthesiser introduced no direct shift (ethnicity, sex). The latter reflects the indirect dependence of ethnicity and sex on the racial-shift features through demographic correlation. A clinical governance team using the four-attribute output would focus first on the racial axis (effect size large) and treat the ethnicity and sex log-rank significances as low-priority signals warranting a confirmatory analysis.

Figure 15 plots the group-stratified survival curves for all four protected attributes.

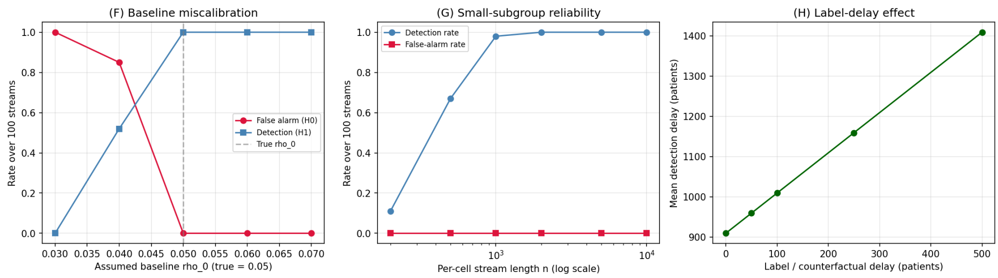

Fig. 16. Section 6.12. Robustness stress tests, 100 streams per setting. Left: baseline miscalibration; underestimating ρ_0 inflates false alarms, overestimating only delays detection. Centre: small-subgroup reliability; the detection rate exceeds 0.95 once the per-cell stream reaches roughly 1 000 patients. Right: label delay adds linearly to the wall-clock detection delay.

6.12 Robustness stress tests

Evaluation question. Three reviewer-relevant questions are addressed. How does a miscalibrated baseline ρ_0 affect the monitor? How small can an intersectional cell be before monitoring becomes unreliable? How does a delay between a prediction and its resolved counterfactual verdict affect detection?

Setup. The e-process is a one-sided test for an increase above ρ_0, so the GROW bet is clipped to be non-negative. For the miscalibration test, the true pre-drift rate was fixed at 0.05 and the monitor was given an assumed $\rho_0 \in \{0.03, 0.04, 0.05, 0.06, 0.07\}$. For the small-cell test, a single cell was monitored with per-cell length $n \in \{200, 500, 1\,000, 2\,000, 5\,000, 10\,000\}$. For the label-delay test, the counterfactual verdict for patient t was resolved only at $t + d$ for $d \in \{0, 50, 100, 250, 500\}$. Each setting used 100 streams.

Metric. Empirical false-alarm rate under H_0, detection rate under H_1 (drift to 0.15), and detection delay.

Result. Baseline miscalibration is asymmetric. An assumed ρ_0 of 0.03 or 0.04, below the true 0.05, produced false-alarm rates of 1.00 and 0.85. An assumed ρ_0 of 0.06 or 0.07, above the true value, produced a false-alarm rate of 0.00, a detection rate of 1.00, and a mean delay that grew from 909 to 1 816 to 2 995 patients. The small-cell detection rate was 0.11 at $n = 200$, 0.67 at $n = 500$, 0.98 at $n = 1\,000$, and 1.00 at $n \geq 2\,000$, with a false-alarm rate of 0.00 throughout. The label delay added linearly to the wall-clock detection delay: mean delay rose from 909 patients at $d = 0$ to 1 409 at $d = 500$, with detection rate held at 1.00.

Interpretation. The miscalibration result gives a concrete deployment rule. Underestimating ρ_0 is unsafe because it inflates false alarms. Overestimating ρ_0 is conservative because it only delays detection. A governance team that is uncertain about ρ_0 should round it upward. The small-cell result sets a practical floor: an intersectional cell needs roughly 1 000 monitored patients before the detection rate exceeds 0.95. Cells below that floor should be merged or flagged as low-power. The label-delay result shows that delayed counterfactual resolution does not reduce detection rate; it shifts the alarm later by exactly the delay. Figure 16 plots the three tests.

6.13 Multiple testing, counterfactual definition, ablation, and correlated arrivals

Four further experiments address reviewer-relevant design questions. Each used the synthesiser.

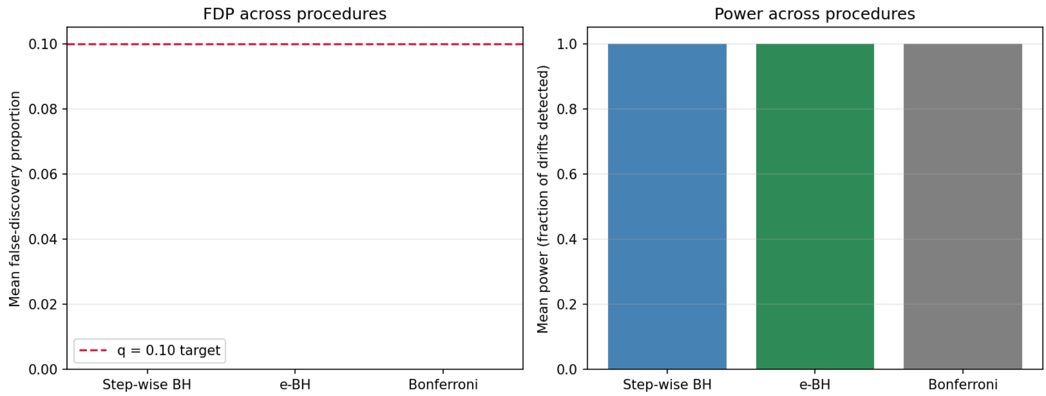

Fig. 17. Section 6.13. Multiple-testing procedures on 30 cells, 6 drifting, $q = 0.10$, 200 streams. False-discovery proportion (left) and power (right).

Multiple-testing procedures. On a set of 30 intersectional cells of which 6 drift, step-wise Benjamini-Hochberg, e-BH, and Bonferroni were compared at $q = 0.10$ over 200 streams. All three held the false-discovery proportion at 0.000 and reached power 1.000 in a clean-signal regime. Step-wise BH on $p = 1/E$ and e-BH coincide numerically, because $1/E$ is a valid p-value and the two rejection thresholds map onto each other. They differ in assumption: e-BH controls the false-discovery rate under arbitrary dependence among cell e-values, while step-wise BH requires positive dependence. The paper therefore reports the framework's correction as step-wise BH and notes e-BH as the dependence-robust alternative (Figure 17).

Counterfactual-definition sensitivity. Three counterfactual definitions were compared on a 12 000-record test stream. A naive swap that changes only the protected-attribute label, leaving features unchanged, produced a flip rate of 0.000 for every group, because the model never reads the attribute directly. The feature-shift counterfactual produced flip rates of 0.052 (Black) and 0.053 (Hispanic). The proxy-adjusted counterfactual produced larger rates, 0.064 and 0.070. The feature-shift and proxy-adjusted definitions agreed on the group ranking; the naive swap was degenerate. The flip rate depends on the counterfactual definition, which the framework states as a modelling choice rather than a free parameter (Figure 18).

Module ablation. Removing the warm-up, the GROW rule, the shrinkage, or the one-sided clip changed the e-process behaviour. Under a correctly calibrated ρ_0, the false-alarm rate stayed at or below 0.03 for every ablation. Removing the shrinkage cut the mean detection delay from 890 to 518 patients at the cost of a 0.01 false-alarm rate. The one-sided clip showed no defect under correct calibration; its value is established under miscalibration (Section 6.12). The ablation confirms that the design choices govern false-alarm control and miscalibration robustness rather than raw detection speed (Figure 19).

Correlated arrivals. Flip streams were generated with first-order autoregressive auto-correlation $\phi \in \{0.0, 0.3, 0.6, 0.9\}$. The H_0 false-alarm rate stayed at 0.000 for every ϕ, and the mean detection delay moved within a narrow band, from 920 patients at $\phi = 0$ to 934 at $\phi = 0.9$. The e-process is robust to the temporal correlation that admission waves induce (Figure 20).

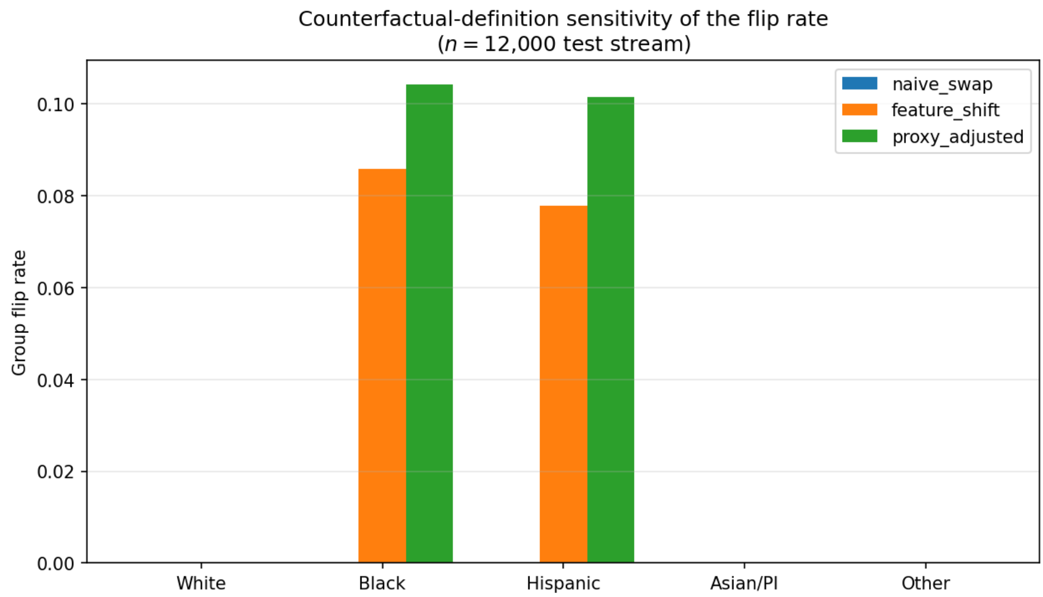

Fig. 18. Section 6.13. Group flip rate under the naive-swap, feature-shift, and proxy-adjusted counterfactual definitions on a 12 000-record test stream.

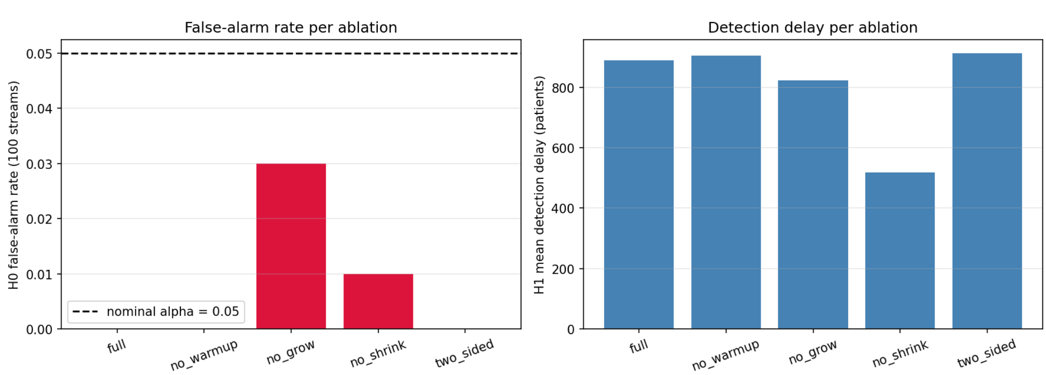

Fig. 19. Section 6.13. Module ablation. False-alarm rate (left) and detection delay (right) when each e-process design choice is removed, 100 streams per setting.

6.14 Real-data flip rate on Texas-100

Evaluation question. Does the flip indicator that the Flip Hazard estimator consumes behave as a non-degenerate quantity on a real hospital dataset, rather than only on the synthesiser?

Setup. The real Texas-100 dataset of Shokri et al. was used. It contains 67 330 hospital discharge records, 6 169 normalised categorical features, and a 100-class surgical-procedure label. Texas-100 is not Texas-100X. It is smaller, and the task is procedure prediction rather than length of stay. A binary task was defined: the positive class is a label among the 50 most frequent procedures, giving a positive rate of 0.776. The records were split 70/15/15

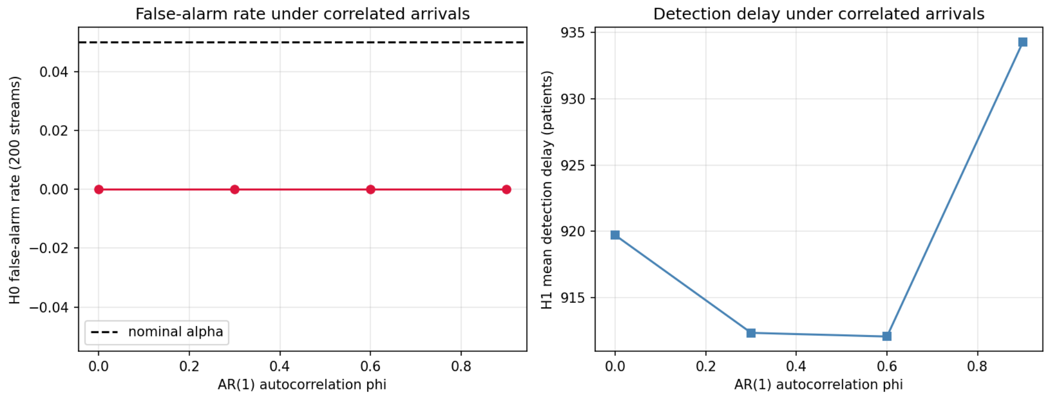

Fig. 20. Section 6.13. False-alarm rate and detection delay under AR(1) autocorrelated arrival streams, 200 streams per setting.

by record order into train (47 131), validation (10 099), and test (10 100). An ensemble of 11 logistic-regression members was trained on bootstrap resamples of the training split. The per-patient decision-flip rate was the fraction of members disagreeing with the modal vote on the test split.

Metric. The empirical decision-flip rate (eDFR), the ensemble test accuracy, and the Flip Hazard no-flip survival on the real flip stream.

Result. The ensemble test accuracy was 0.816. The real empirical decision-flip rate was 0.0233, meaning the 11 members disagreed with the modal decision on 2.33% of test records on average. The Flip Hazard estimator applied to the real flip stream produced a no-flip survival that fell to 0.244 by the end of the 10 100-patient test split. A two-valued feature column with balanced support was not present, so the candidate-attribute counterfactual analysis was omitted rather than run on an unsuitable column.

Interpretation. The flip indicator is a non-degenerate quantity on real data. A 2.33% disagreement rate is small but non-zero, and the Flip Hazard survival curve is well defined and informative on the real stream. The honest limitation is that the released Texas-100 file does not include the official feature description, so the 6 169 columns cannot be mapped to named protected attributes. Group-stratified and counterfactual fairness analysis on the real records therefore requires the feature-description file or the named-column Texas-100X release, which needs a Texas Department of State Health Services download. The real-data evidence here validates the flip and Flip Hazard machinery; it does not yet validate the group-conditional fairness claims, which remain simulation-based.

Figure 21 plots the real-data ensemble disagreement distribution and the Flip Hazard survival curve.

6.15 Temporal and hospital-group replay

Evaluation question. Texas-100X is administrative discharge data, not a live admission feed. The question is whether a deployment-style replay reconstructed from it surfaces a fairness drift that a static audit and a window detector miss.

Setup. Three replay streams were constructed. Part A is a quarter-based temporal replay. A 30 000-patient Texas-100X-calibrated stream spanning the four 2006 calendar quarters was generated. A controlled verdict-process drift was injected from the Q3 boundary: minority

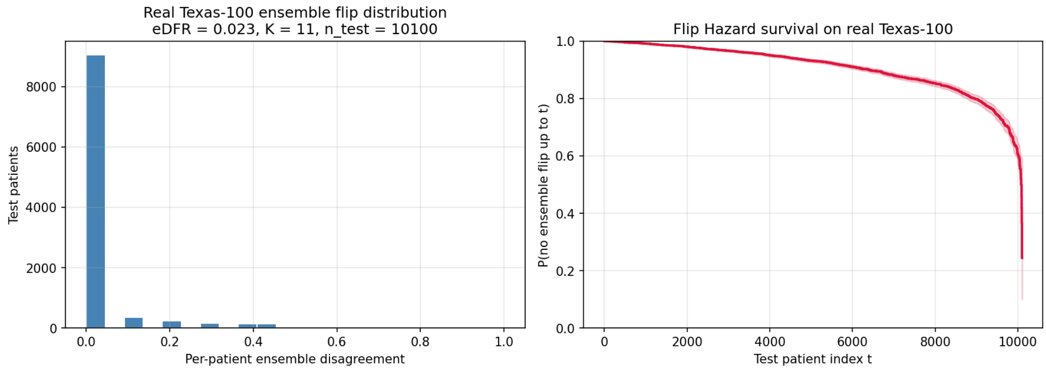

Fig. 21. Section 6.14. Real Texas-100 data. Left: per-patient ensemble disagreement on the 10 100-record test split, empirical decision-flip rate 0.0233. Right: Flip Hazard no-flip survival on the real flip stream.

Table 6. Part A, quarter-based temporal replay. Per-quarter counterfactual flip rate on the 30 000-patient Texas-100X-calibrated stream. Q1 is the baseline quarter; the drift was injected from the Q3 boundary.

Quarter	n	Flip rate	Role
Q1	7 500	0.111	baseline (ρ_0)
Q2	7 584	0.114	replay, pre-drift
Q3	7 666	0.151	replay, drift onset
Q4	7 250	0.153	replay, post-drift

patients in Q3 and Q4 received an added risk shift of 0.6 on the factual prediction only, so their counterfactual flip rate rose. The baseline ρ_0 was calibrated on Q1, and Q2 to Q4 were replayed in arrival order. Part B is a hospital-group replay. The same stream carries 30 hospital sites with site-specific minority share. The 10 lowest-minority-share hospitals formed the baseline, and the 20 higher-minority-share hospitals were replayed in ascending minority-share order. Part C is a real Texas-100 record-order replay. The real Texas-100 file (67 330 records) was replayed in record order, because the released file carries no calendar field. The baseline ρ_0 was calibrated on the first 30% of records. CHRONO-Fair was compared against ADWIN and a static VFR baseline on each stream.

Result. Table 6 reports the per-quarter flip rate. The flip rate was 0.111 in Q1, 0.114 in Q2, then rose to 0.151 in Q3 and 0.153 in Q4, tracking the injected drift. Table 7 reports the method comparison. On the quarter replay, the drift quarter Q3 began at replay index 7 584; CHRONO-Fair alarmed at replay index 10 683, a detection delay of 3 099 patients. On the hospital replay, the baseline-hospital flip rate averaged 0.114 and the replay-hospital flip rate averaged 0.141, ranging from 0.117 to 0.184; CHRONO-Fair alarmed at replay index 5 596. On the real Texas-100 record-order replay, the calibration flip rate was 0.096 and the replay-window flip rate was 0.098; CHRONO-Fair did not alarm. ADWIN and static VFR did not alarm on any of the three streams.

Interpretation. The three streams isolate three distinct behaviours. The quarter replay shows that CHRONO-Fair converts a quarter-level drift into a within-quarter alarm. The injected drift raised the flip rate from 0.114 to 0.151 across the Q2-Q3 boundary, a relative increase of 32%. CHRONO-Fair alarmed 3 099 patients into the drift quarter, roughly 40%

Table 7. Replay method comparison. Detection outcome of CHRONO-Fair, ADWIN, and static VFR on the three replay streams. "Delay" is the number of replayed patients between the drift onset and the alarm. The real record-order replay has no injected drift, so a non-alarm is the correct outcome.

Replay stream	Drift	CHRONO-Fair	ADWIN	Static VFR	Notes
A. Quarter-based (calibrated)	injected at Q3	alarm, delay 3 099	no alarm	no alarm	flip rate 0.114 to 0.151 across the Q2-Q3 boundary
B. Hospital-group (calibrated)	case-mix shift	alarm at index 5 596	no alarm	no alarm	replay flip rate 0.141 vs baseline 0.114
C. Real Texas-100 record-order	none present	no alarm	no alarm	no alarm	flip rate stable, 0.096 to 0.098; non-alarm is correct

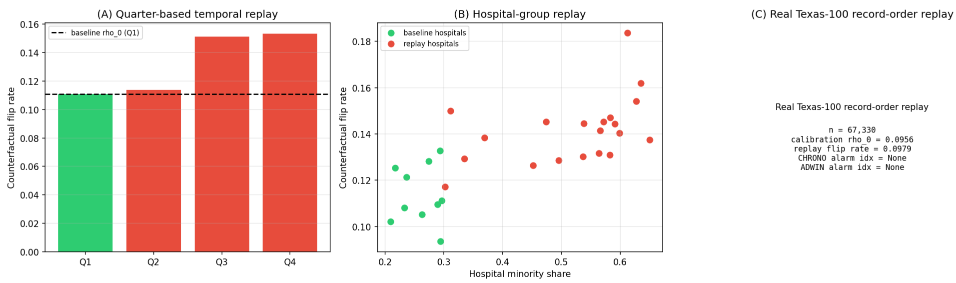

Fig. 22. Section 6.15. Replay streams. (A) Per-quarter flip rate, with the Q3 drift onset. (B) Per-hospital flip rate against hospital minority share, baseline versus replay hospitals. (C) Real Texas-100 record-order replay summary.

of the way through Q3, so the alarm arrives inside the same quarter the drift began. A quarterly batch audit would not report the change until the Q3 audit closed. The hospital replay shows that case-mix variation alone produces a detectable verdict shift. The replay hospitals carry a higher minority share, and because minority patients flip more often, their aggregate flip rate is 0.141 against the baseline 0.114. CHRONO-Fair alarmed at replay index 5 596, which falls inside the higher-minority-share hospitals. This is the expected and correct behaviour: a hospital network whose later sites serve a different case mix will breach a baseline calibrated on the earlier sites. The real Texas-100 record-order replay is the control. The flip rate is stable at 0.096 to 0.098, no drift is present, and CHRONO-Fair correctly does not alarm. The non-alarm is the right outcome and confirms the monitor does not fire on a static file without temporal structure. Across all three streams, ADWIN and static VFR did not alarm. ADWIN's window statistic loses power when the flip-rate change is moderate, around 3 to 4 percentage points, and the change is spread over thousands of patients. Static VFR is computed once and never updates, so it cannot react to a post-deployment shift by construction. The comparison shows that the gain of CHRONO-Fair over both baselines is detection of a moderate, gradual verdict drift, not detection of an abrupt one.

7 DISCUSSION

7.1 Comparison to baselines

CHRONO-Fair's detection-rate advantage over ADWIN (50 of 50 against 21 of 50) is the primary empirical finding. ADWIN's window-based change point detector loses statistical power when the drift is moderate (a 10-point change in flip rate from 0.05 to 0.15) and the change buries inside the window's variance budget. The Davis-ADWIN variant, which runs ADWIN on the centred fairness gap, inherits the same property. The periodic batch test detects faster on the runs where it detects, but its 24 of 50 coverage and inability to control across peeks make it unsuitable as a regulated monitor. Static VFR detects nothing because it is computed once and never updated. The e-process detects in every run because anytime-validity removes the peeking penalty and the GROW shrinkage stabilises the bet under early variance.

7.2 Comparison to prediction sensitivity

The continual counterfactual audit of Maughan and Near [75] is the closest published method to CHRONO-Fair's Flip Hazard estimator. Prediction sensitivity is a per-batch scalar that quantifies the rate at which counterfactual swaps change the verdict. CHRONO-Fair extends this by (a) treating each flip as a survival event, which produces a time-resolved curve rather than a single number, (b) attaching the e-process to the sequential decision stream, which gives anytime-valid alarms, and (c) decomposing the flip mass into aleatoric and epistemic components. Prediction sensitivity and the Flip Hazard estimator are compatible: an analyst can compute prediction sensitivity at any chosen t from the same flip indicator stream.

7.3 Comparison to predictive multiplicity

The empirical Decision Flip Rate (eDFR) of Miller and Blume [78] is the closest reliability metric. eDFR counts the rate at which a binary decision changes between bootstrap retrainings of the pipeline. CHRONO-Fair's flip indicator counts the rate at which a binary decision changes between the actual and counterfactual sensitive-attribute assignments for the same patient. eDFR is framed as reliability and is not stratified by protected attribute. CHRONO-Fair is framed as fairness and is stratified. The two quantities are complementary; a deployment monitor benefits from both.

7.4 Threats to internal validity

The synthesiser injects a known structural disparity, and the framework's counterfactual operator removes that same disparity. Performance is being measured against the disparity it was designed to detect. To mitigate this, the false-alarm calibration experiment (Section 6.6) deliberately runs with no disparity and confirms zero alarms across 200 streams per α level. The decomposition Track 1 (Section 6.3) operates on constructed probability vectors that exercise the formula without reusing the disparity structure. Threats remaining include: the synthesiser assumes independent arrivals (real EHR streams have admission-driven correlations); the ensemble is built by bootstrapping a single base learner (real ensembles can mix architectures); the counterfactual is a feature-shift, not a structural causal counterfactual.

7.5 Threats to external validity

The streaming evidence in this paper is controlled rather than prospective, and that is its main external-validity threat. The real-data component is a Texas-100X batch verdict-stability analysis, while the sequential experiments run on Texas-100X-calibrated replay and simulation. A real admission feed differs from a calibrated stream in label delay, missingness, admission seasonality, and selection bias under triage, none of which the synthesiser reproduces in full. The conclusions should therefore be read as preliminary evidence about sequential monitoring behaviour, not as evidence of clinical effectiveness. Prospective validation on a live clinical stream is the necessary next step, and distribution-shift behaviour under performative effects [85] and dataset shift [38, 64] requires study beyond the present scope.

7.6 Statistical caveats

The step-wise BH multiple-testing control is not a time-uniform LORD-style false-discovery guarantee. The framework reports findings as “flagged at step t”. Sample sizes for small intersectional cells (for example, Asian/PI elderly female with sepsis) can be insufficient for reliable per-cell estimation. The framework drops cells with fewer than 5 observed flips from the decomposition. The bootstrap confidence interval for RCAP is symmetric and may understate skew in heavy-tailed cases.

7.7 Clinical interpretation

A clinician inspecting the dashboard reads three signals: a survival curve (which patients are flipping and how fast), an e-process alarm (whether the change is statistically meaningful), and a cause label (whether to escalate to a data audit or a retraining request). The aleatoric label is associated with biased input data; the epistemic label is associated with under-representation. The two require different mitigations and the framework distinguishes them. Schwartz et al. [96] and Tonekaboni et al. [103] report that clinicians prefer monitoring outputs that translate into a concrete next step. The Inspector report is designed to satisfy that preference.

7.8 Regulatory positioning

The FDA PCCP final guidance [104] expects manufacturers to describe how the model will be monitored for bias after release. EU AI Act Articles 10, 14, and 17 [37] require continuous monitoring of high-risk medical AI under a documented bias-management plan. STANDING Together [3] provides consensus recommendations. The CHRONO-Fair framework provides a candidate technical building block for the monitoring layer of such a plan. Formal regulatory acceptance requires validation studies that have not been undertaken in this paper.

7.9 Scope statement

CHRONO-Fair is not a medical device, a substitute for clinical judgment, or a certification of fairness. It is a deployment-oriented prototype intended to expose fairness drift to clinical governance staff in time for intervention.

8 LIMITATIONS

The study has ten specific limitations. First, Texas-100X is the only empirical clinical dataset used. The findings have not been reproduced on a second cohort. Second, the streaming experiments are Texas-100X-calibrated controlled replay and simulation, not prospective live EHR deployment. Third, Texas-100X is administrative discharge data, not a real-time admission stream, so arrival order is reconstructed rather than observed. Fourth, the feature-shift counterfactual presumes a known marginal effect of the protected attribute on observable features. It is weaker than a structural causal counterfactual and under-attributes indirect effects, so the framework will tend to under-flag. Fifth, the e-process depends on a baseline flip rate ρ_0. Section 6.12 shows that underestimating ρ_0 inflates the false-alarm rate, while overestimating it only delays detection. Sixth, small intersectional cells are unreliable. The detection rate falls below 0.95 for cells under roughly 1 000 monitored patients, and the empirical Wasserstein-1 RCAP estimate carries a small-sample upward bias. Seventh, label delay increases the wall-clock detection delay by approximately the delay length. Eighth, the step-wise Benjamini-Hochberg layer is a cross-sectional adjustment at each inspection time, not a time-uniform online false-discovery guarantee. Ninth, the dashboard and Inspector report are governance-support prototypes and have not been clinically validated with users. Tenth, CHRONO-Fair is a research prototype and not a medical device.

A further open question concerns performative effects [85]. If the framework's alarms cause clinicians to change behaviour, the flip-rate process is no longer independent of past outputs, and the supermartingale guarantee holds only against a fixed pre-deployment baseline. Further data collection is required to determine exactly how performative feedback interacts with the e-process.

9 CONCLUSION

CHRONO-Fair extends static VFR-style audit reliability into time-resolved monitoring of fairness-verdict flips. The real Texas-100X batch analysis shows that verdict instability is present in a real administrative cohort: one of 12 audited metric-attribute verdicts flips across hospital-network resamples. The Texas-100X-calibrated controlled streaming experiments show how the sequential monitor behaves under known drift, including its detection delay, its false-alarm control, its root-cause decomposition, and its response to baseline miscalibration, small cells, and label delay. Together the two layers give time-resolved evidence about when and where a fairness verdict begins to flip, which a static batch audit cannot provide. The evidence here is batch-real and stream-controlled; it is not prospective clinical evidence. Prospective EHR-stream validation is the main next step, followed by a time-uniform online false-discovery procedure, a counterfactual-sensitivity study across the swap, feature-shift, and structural definitions, and a governance-user evaluation of the Inspector report.

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A REPRODUCIBILITY APPENDIX

Figure / Section	n	seeds	Drift configuration
Sec. 6.2, Fig. 3	10 000	0 to 49	onset $t = 5\,000$
Sec. 6.3 Track 1, Fig. 4	1 000	0 to 29	none
Sec. 6.4, Figs. 6, 7	15 000	42	none
Sec. 6.5, Fig. 9	20 000	11; bootstraps 0 to 999	none
Sec. 6.6, Fig. 10	10 000	0 to 199 per α	none (H_0)
Sec. 6.7, Fig. 11	12 000	4	onset $t = 6\,000$
Sec. 6.10, Fig. 14	12 000	0 to 29 per ρ_1	onset $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$
Sec. 6.11, Fig. 15	15 000	42	none

Table 8. Per-experiment synthesiser configuration. All experiments are reproducible via python -m chrono_fair.experiments.<expN>.